

Denis Diderot
D'Alembert's Dream
1769

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D'Alembert's Dream (*Le Rêve D'Alembert*) is the title traditionally given to three related dialogues written by Diderot in 1769 but not published until after his death.

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Denis Diderot
A Conversation Between D'Alembert and Diderot

D'ALEMBERT

I grant that a Being which exists somewhere yet corresponds to no single point in space, a Being which has no spatial dimensions yet occupies space, which is wholly complete in each part of this space, which is essentially different from material stuff yet is united with it, which is affected by matter and moves it without moving itself, which acts on matter yet is subject to all its changes, a Being about which I haven't the slightest conception—a Being with such a contradictory nature is difficult to accept. But there are other obscurities waiting for anyone who rejects such a Being. For in the last analysis, if this sensitivity which you put in its place is a universal and essential quality of matter, then a stone must feel.¹

DIDEROT

Well, why not?

D'ALEMBERT

That's hard to believe.

DIDEROT

Yes, for the man who cuts the stone, carves, and grinds it without hearing it cry out.

D'ALEMBERT

I really wish you'd tell me what difference you establish between a man and a statue, between marble and flesh.

DIDEROT

Not much. One can make marble with flesh and flesh with marble.

D'ALEMBERT

But one is not the other.

DIDEROT

Just the way what you call living energy is not the same as latent energy.

D'ALEMBERT

I don't understand you.

¹Diderot's materialistic view of creation sees sensitivity (or sensation) actually or potentially present in all things, animate and inanimate.

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AND DIDEROT

DIDEROT

Let me explain. The transporting of a body from one place to another is not motion—it's only its effect. Motion is equally present both in the transported body and in the one which remains motionless.

D'ALEMBERT

That way of seeing it is new.

DIDEROT

Nonetheless, it's true. If you remove the obstacle which prevents the local movement of a stationary body, it will be shifted. If by sudden rarefaction you get rid of the air which surrounds the enormous trunk of this oak tree, then the water it contains will suddenly expand and blow it up into a hundred thousand fragments. I'm saying the same thing is true for your own body.

D'ALEMBERT

All right. But what's the relationship between motion and sensitivity? Could it by chance be the case that you recognize an active sensitivity and a latent sensitivity, just as there is an active and latent force? An active force which manifests itself by displacement, and a latent force which manifests itself by pressure, an active sensitivity which is characterized by certain observable actions in an animal and perhaps in a plant, and a latent sensitivity which we would confirm by its transformation into a condition of active sensitivity.

DIDEROT

Splendid. You've got it.

D'ALEMBERT

Thus, the statue only has a latent sensitivity, and man, animals, and perhaps even plants are endowed with an active sensitivity.

DIDEROT

There's no doubt about this difference between the block of marble and fleshy tissue. But you do understand that that's not the only difference.

D'ALEMBERT

Of course. Whatever resemblance there may be between the exterior form of the man and the statue, there is no connection between their internal organic structures. The chisel of the most expert sculptor can't make even an epidermis. But there's an extremely simple process for making latent energy transform

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itself to active energy. It's an experiment which is repeated a hundred times a day right before our eyes; whereas, I don't see how one can make a body move from a state of latent sensitivity into a state of active sensitivity.

DIDEROT

That's because you don't want to see it. The phenomenon is common enough.

D'ALEMBERT

Please tell me what this common enough phenomenon is.

DIDEROT

I'll tell you because you don't mind the shame of being told. It happens every time you eat.

D'ALEMBERT

Every time I eat!

DIDEROT: Yes, because when you're eating, what are you doing?

You're removing the obstacles which stand in the way of the active sensitivity of what you're eating. You assimilate the food into yourself. You make flesh out of it. You turn it into animal stuff. You make it capable of sensation. And what you do to food, I'll do to marble whenever I like.

D'ALEMBERT

And how will you do that?

DIDEROT

How? I'll make it edible.

D'ALEMBERT

Make marble edible—that doesn't seem easy to me.

DIDEROT

It's up to me to show you how it's done. I take the statue which you see. I put it into a mortar and with some heavy blows with a pestle . . .

D'ALEMBERT:

Please go gently—it's a masterpiece by Falconet. If it were a piece by Huez or someone else . . .

DIDEROT

That means nothing to Falconet. The statue's paid for, and

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AND DIDEROT

Falconet doesn't care much what people think of him now and not at all about his future reputation.²

D'ALEMBERT

All right then, pulverize away.

DIDEROT

When the block of marble has been reduced to a very fine powder, I mix this powder with some humus or topsoil. I knead them together well. I water the mixture and let it rot for a year, two years, a century—the time doesn't matter. When it's all been transformed into almost homogeneous matter, into humus, do you know what I do?

D'ALEMBERT

I'm sure you won't be eating the humus.

DIDEROT

No, but there is a way of uniting that humus and myself, of appropriating it—a *latus*, as the chemist would say.

D'ALEMBERT

And this *latus* is a plant?

DIDEROT

Very good. I sow some peas, beans, cabbages, and other leguminous plants in it. The plants nourish themselves on the earth, and I nourish myself on the plants.

D'ALEMBERT

True or false, I like this transformation of marble into humus and of humus into the vegetable realm and of the vegetable realm into the animal realm, into flesh.

DIDEROT

In this way I make flesh or soul, as my daughter says—active, sensitive matter. And if I have not resolved the problem which you proposed to me, at least I have come a great deal closer. For you will concede to me that the distance from a piece of marble to a sentient being is a lot further than from a sentient being to a thinking being.

² Étienne Falconet (1716-1791): a leading French sculptor. Jean-Baptiste d'Huez (1729-1793): a well-known French sculptor.

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D'ALEMBERT

I agree. But for all that a sentient being is still not the same as a thinking being.

DIDEROT

Before taking another pace forward, let me give you the history of one of the greatest mathematicians in Europe. What was he at first, this marvelous being? Nothing at all.

D'ALEMBERT: How could he be nothing? Nothing comes from nothing.

DIDEROT: You're taking the words too literally. What I mean is that before his mother, the beautiful and scandalous Canoness Tencin, had attained the age of puberty and before the soldier La Touche was an adolescent, the molecules which were to form the first rudiments of my mathematician were scattered in the young and immature mechanical structures of both of them. They were filtered with the lymph and circulated with the blood until finally they settled in the reservoirs destined for their union, the sex glands of his mother and his father. And there this rare germ is formed, and there, according to common opinion, it's led along the fallopian tube into the womb, attaches itself to the womb by a long peduncle, and there grows in stages and develops into the fetal state. Then comes the moment when it leaves its dark prison, and, behold, he is born, exposed on the steps of Saint-Jean-le-Rond, which gave him his name. Taken from the Foundling's Home, set on the breast of the good wife of a glazier, Madame Rousseau, nursed, grown large in body and mind, he becomes a writer, engineer, and mathematician. How did that all happen? By eating and other purely mechanical operations. Here in four words is the general formula: eat, digest, distil in *vasti licito, et fiat homo secundum artem* [*in the proper container, and let a man be made by the usual art*]. And anyone who explained to the Academy the progress of the formation of a man or an animal would only have to refer to material agents whose successive effects would be an inert being, a sentient being, a thinking being, a being solving the problem of the precession of the equinoxes, a sublime being, a marvelous being, a being who grows older, declines, dies, dissolves, and is returned to vegetative earth.³

³The details of this "being" come from d'Alembert's life. He was the illegitimate child of Claudine de Tencin (who had entered a convent and later left) and a military officer Louis-Camus Destouches (whom Diderot calls La Touche). Shortly after his birth, his mother abandoned him outside the church mentioned. D'Alembert went into a foundlings' home and was later adopted. His father continued to provide generous support for his son's education. D'Alembert took

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D'ALEMBERT

So you don't believe in pre-existing germ cells?

DIDEROT

No.

D'ALEMBERT

Ah, I'm pleased to hear that.

DIDEROT

It's against experience and reason. It contradicts the experience of anyone who could have wasted his time looking for these germ cells in the egg and in most animals before a certain age, and it contradicts reason which teaches us that the divisibility of matter has a natural limit, although there is no limit to such divisibility in our understanding, and which rejects the idea of an elephant completely formed in an atom and in this atom another fully shaped elephant, and so on *ad infinitum*.⁴

D'ALEMBERT

But without these pre-existing germ cells, the original generation of animals cannot be imagined.

DIDEROT

If you're troubled by the question of whether the egg came before the hen or the hen before the egg, that's because you assume that animals were originally what they are now. What foolishness! We don't know any more about what they were than we do about what they'll become. The imperceptible earthworm which moves around in the mud is perhaps in the process of developing into a large animal, and an enormous animal, which astonishes us with its size, is perhaps in the process of developing into an earthworm and is perhaps a unique and momentary production of this planet.

D'ALEMBERT

What did you mean by that?

the name Jean Le Rond d'Alembert after the church where he was abandoned (Saint-Jean-le-Rond) and went on to become an outstanding and celebrated mathematician and a leading figure in the Enlightenment. The collection of special cells from all over body in the sexual organs and the "blending" of those cells from the father and mother in the offspring was one of a number of theories of inheritance at the time. The precession of the equinoxes is a slow change in the orientation of the rotational axis of the earth (like the wobble on a top). D'Alembert refined the equations used to explain the phenomenon.

⁴The "preformation" hypothesis held that the reproductive material consists of an infinite series of tiny adult forms one inside another (like a collection of Russian wooden dolls).

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DIDEROT

I was telling you . . . But that will make us digress from our original discussion.

D'ALEMBERT

What does that matter? We'll come back to it or we won't.

DIDEROT

Will you allow me to leap ahead a few thousand years?

D'ALEMBERT

Why not? Time means nothing to nature.

DIDEROT

You'll consent that I extinguish our sun?

D'ALEMBERT

Happily—all the more so because it won't be the first one to be extinguished.

DIDEROT

Once the sun goes out, what will happen? The plants will die, the animals will die, and there we have the earth—lonely and quiet. Relight this star, and right away you re-establish the necessary cause of an infinite number of new generations, and I wouldn't venture to guarantee that with the succession of ages our plants and animals of today would be reproduced or not reproduced among these new generations.

D'ALEMBERT

Why wouldn't the same scattered elements, once they start to reunite, bring back the same results?

DIDEROT

Because everything in nature is interconnected, and the man who assumes a new phenomenon or brings back a moment from the past is creating a new world once again.

D'ALEMBERT

That's something a profound thinker could not deny. But to return to man, since the universal order wished him to exist. You recall you left me in that transition from a sentient being to a thinking being?

DIDEROT

I remember.

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D'ALEMBERT

To be frank, you'd do me a great favour to take me beyond that.
I'm eager to start thinking.

DIDEROT

But if I didn't manage to succeed what would result, given the
sequence of incontestable facts?

D'ALEMBERT

Nothing, other than that we'd be stopped in our tracks at that
point.

DIDEROT

And to move on further, would we be permitted to invent an
agent with contradictory attributes, a word without meaning and
unintelligible?

D'ALEMBERT

No.

DIDEROT

Could you tell me what the existence of a sentient being is in
relation to itself?

D'ALEMBERT

It's the consciousness of having been itself from the first
moment of reflection until the present moment.

DIDEROT

And what is this consciousness founded on?

D'ALEMBERT: On the memory of its actions.

DIDEROT

And without this memory?

D'ALEMBERT

Without this memory there'd be no "itself." Since it would not
sense its existence except at the moment of receiving an
impression, it would have no history of its life. Its life would be
an interrupted series of sensations in which nothing was
connected.

DIDEROT

Very good. Now, what is memory? Where does it arise?

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D'ALEMBERT

From a certain organic structure which grows, diminishes, and sometimes is lost completely.

DIDEROT

So then, if a being which feels and has this organic structure suitable for memory links together the impressions it receives and forms by this linking together a history of its life and acquires a consciousness of itself, then it denies, it affirms, it concludes—it thinks.

D'ALEMBERT

That's what it seems to me. I have only one remaining difficulty.

DIDEROT

You're wrong. You still have a lot more difficulties.

D'ALEMBERT

But one main one. It strikes me that we can think about only one thing at a time and in order to form a simple proposition (I'm not talking about those enormous chains of reasoning which include thousands of ideas in their development), we would say that it's necessary to have at least two things present, the object which seems to sit there under the eye of our understanding, while our understanding is busy with the quality which it will affirm or deny about that object.

DIDEROT

I share that concern. And it's led me sometimes to compare our organic fibres with sensitive vibrating strings. A sensitive vibrating string still oscillates and resonates a long time after one has plucked it. It's this oscillation, this sort of inevitable resonance, which holds the present object, while our understanding is busy with the quality which is appropriate to it. But vibrating strings have yet another property—they make other strings quiver. And thus an initial idea summons up a second, and these two a third, then all three a fourth, and so it goes, without our being able to set a limit to the ideas which are aroused and linked in a philosopher who meditates or listens to himself in silence and darkness. This instrument makes astonishing leaps, and one recalled idea sometimes is going to set in motion a harmonic at an incomprehensible interval. If the phenomenon is perceptible between resonating strings, inert and separated, how would it not take place between vital points linked together, between continuous and sensitive fibres?

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D'ALEMBERT

If that's not true, it's at least very ingenious. But one would be tempted to think that you're falling imperceptibly into a difficulty which you wished to avoid.

DIDEROT

What's that?

D'ALEMBERT

You object to the distinction between the two substances, matter and spirit?⁵

DIDEROT

I don't hide the fact.

D'ALEMBERT

Yet if you look closely at this idea, you're making the understanding of a philosopher an entity distinct from the instrument, a sort of musician who presses his ear against the vibrating strings and makes judgments about their consonance or dissonance.

DIDEROT

It's possible I have prompted this objection. But perhaps you'd not have made it if you'd considered the difference between the philosophical instrument and the instrumental harpsichord. The philosophical instrument is sentient—he is at the same time the musician and the instrument. As something sentient he has the momentary consciousness of the sound he is making; as an animal, he has the memory of that. This organic faculty, by linking the sounds he has within him, produces and keeps the melody there. Suppose there is a harpsichord with sensitivity and a memory. Tell me if it won't know and repeat on its own the melodies you have executed on its keys. We are instruments endowed with sensibility and memory. Our senses are so many keys which are struck by nature surrounding us and which often strike themselves. And there we have, in my judgment, everything which goes on in an organic harpsichord like you and me. There's an impression that has its cause either inside or outside the instrument, a sensation which is born from this impression, a sensation which lasts, for it is impossible to imagine that it is made and extinguishes itself in an indivisible instant, another impression which follows this one, and which

⁵Diderot's text says "distinction between two substances," a reference to the belief that mind and matter are fundamentally different (as Descartes maintained). Diderot is committed to a thoroughgoing materialism. I have added "matter and spirit" to make this point clear.

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similarly has its cause either inside or outside the animal, a second sensation and voices which designate them by natural or conventional sounds.

D'ALEMBERT

I see. And so if this sentient and animated harpsichord was now endowed with the faculty of feeding and reproducing itself, it would live and, either on its own or with its female partner, give birth to little keyboards, living and resonating.

DIDEROT

No doubt. In your view, is a lark, a nightingale, a musician, or a man anything else? And what other difference do you find between a canary and a canary music box?⁶ You see this egg? That's what enables us to overturn all the schools of theology and all temples on the earth. What is this egg? An insensible mass before the germ cell is introduced into it, and after the germ cell is introduced, what is it still? An insensible mass. For this germ cell itself is nothing but an inert and basic fluid. How will this mass develop into a different organic structure, into sensibility and life? By heat. What will produce heat in it? Movement. What will be the successive effects of movement? Instead of answering me, sit down, and let's follow these effects with our eyes from moment to moment. First there's a point which oscillates, a thread which grows and takes on colour, flesh forms, a beak, the tips of wings, eyes, and feet appear, a yellowish material which unwinds and forms intestines. It's an animal. This animal moves, agitates itself, cries. I hear these cries through the egg shell. It is covered with down. It sees. The weight of its head, which moves back and forth, constantly brings its beak against the inner surface of its prison. And then it breaks the shell. It comes out, it walks, it flies, it responds to a stimulus, it runs off, it comes closer, it complains, suffers, loves, desires, rejoices. It has all your moods and goes through all your actions. Do you claim, with Descartes, that this is a purely imitative machine?⁷ But small children will make fun of you, and philosophers will reply that if that's a machine, then you are another machine. If you admit that between you and the animal

⁶Diderot's word for "canary music box" is *serinette*, a small barrel organ used to teach canaries to sing certain tunes. René Descartes (1596-1650), the celebrated and influential French philosopher, maintained that mind and soul were distinct from body and that animals were without souls or mind and were merely mechanical.

⁷Descartes' writings established the famous mind-body problem: if mind is essentially different from matter then where does it come from if all the rest of the world is material stuff and how can mind and matter interact if they are fundamentally different (for it is clear that they do interact). Diderot is addressing this problem by denying that mind and matter are different.

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there is merely a difference in organic structure, you'll be following good sense and reason, acting in good faith. But people will conclude from all this, in opposition to you, that from an inert material arranged in a certain manner, impregnated with another inert material, and subject to heat and movement, we get sensibility, life, memory, consciousness, passions, and thought. There are only two positions you can take. You can imagine that in the inert mass of the egg there is a hidden element which is waiting for the egg to develop in order to manifest its presence, or you can assume that this imperceptible element has insinuated itself into the egg through the shell at a time determined by the developmental process. But what is this element? Did it occupy space or not? How did it come or escape without moving? Where was it? What was it doing there or somewhere else? Was it created at the necessary moment? Was it already in existence waiting for a home? Was it the same stuff as this home or different? If it was the same, then it was material stuff. If it was different one cannot conceive of its inertia before the development of the egg or of its energy in the developed animal. Listen to yourself, and you'll be ashamed. You'll feel that, in order not to admit a simple assumption which explains everything—sensibility as a universal property of matter or a product of organic structure—you're rejecting common sense and jumping into an abyss of mysteries, contradictions, and absurdities.

D'ALEMBERT

An assumption! You're happy to say that. But what if it was a quality essentially incompatible with matter?

DIDEROT

And where do you get the idea that sensibility is essentially incompatible with matter when you don't know the essence of anything, either of matter or of sensitivity? Do you understand better the nature of movement, its existence in a body, and its communication from one body to another?

D'ALEMBERT

Without understanding the nature of sensitivity or that of matter, I see that sensitivity is a simple quality, unified, indivisible, and incompatible with a divisible object or substrate.

DIDEROT

That's metaphysical and theological mumbo jumbo. What? Don't you see that all qualities, all the sensible forms which make up matter are essentially indivisible? There's neither more nor less impenetrability. There is half a round body, but not half

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roundness. There is more or less motion, but movement is neither more nor less—it's either there or it isn't. There's no such thing as a half or a third or a quarter of a head or an ear or a finger, any more than there is a half, a third, or a quarter of a thought. If in the universe there is not a single molecule which resembles another and in a molecule no point which resembles any other point, admit that the atom itself is endowed with a quality, an indivisible form. Concede that division is incompatible with the essential quality of forms, because it destroys them. Be a physical scientist and admit the production of an effect when you see it produced, even though you can't explain to yourself the connection between the cause and the effect. Be logical and don't substitute for a cause which exists and which explains everything another cause which cannot be conceived and whose connection with the effect is even harder to understand, something which produces an infinite multitude of difficulties and which solves none of them.

D'ALEMBERT

Well, what if I give up this cause?

DIDEROT

Then there's only one substance in the universe, in man and in animals. The canary musical box is made of wood, and man is made of flesh. The canary is made of flesh, and the musician is made of flesh organized differently, but the two of them have the same origin, the same formation, the same functions, and the same end.

D'ALEMBERT

And how is the convention of sounds established with your two harpsichords?

DIDEROT

Since an animal is a sensing instrument perfectly similar to another, endowed with the same pattern, equipped with the same strings, plucked in the same manner by joy, sorrow, hunger, thirst, colic, admiration, and terror, it cannot make different sounds at the pole and the equator. That's why you'll find in all languages, living or dead, that interjections are almost the same. We have to derive the origin of all conventional sounds from need and proximity. The sensitive instrument or animal has learned from experience that when it emits a certain sound, there then follows some effect outside itself, that other sensing instruments similar to it or other related animals came close, went away, asked for something, offered something, injured, or caressed it, and these effects were linked in its

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memory and in that of the others to the formation of these sounds. Observe that in human intercourse there are only sounds and actions. And to concede the full strength of my system, notice also that it is subject to the same insurmountable difficulty that Berkeley proposed in arguing against the existence of material bodies.⁸ There is a moment of delirium when the sensitive keyboard thought it was the only keyboard in the world and that all harmony in the universe was coming from it all by itself.

D'ALEMBERT

There's plenty to talk about there.

DIDEROT

That's true.

D'ALEMBERT

For example, we don't understand very well in your system how we form syllogisms or how we draw conclusions.

DIDEROT

But we don't draw them at all. They are all derived by nature. The only thing we do is describe conjoined phenomena whose connection is either necessary or contingent, phenomena which we have learned by experience, necessary in mathematics, physics, and other rigorous sciences, and contingent in morality, politics, and other conjectural sciences.

D'ALEMBERT

Is it true that the connection between phenomena is less necessary in one case than in another?

DIDEROT

No, but the cause is subject to too many particular vicissitudes which escape us, so that we cannot inevitably count on the effect which will follow. The certainty we have that a violent man will lose his temper when he's insulted is not the same as our certainty that a body which strikes a smaller body will set it in motion.

D'ALEMBERT

What about analogies?

⁸ George Berkeley (1685-1753), an Irish bishop, proposed that since minds could not have direct contact with things but only with ideas of things, nothing in the world exists outside our own minds. In order to exist, things must be perceived.

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DIDEROT

An analogy in the most complex cases is only a three-part rule which takes place in the sensing instrument. If some phenomenon known in nature is followed by some other phenomenon known in nature, what will be the fourth phenomenon which follows from a third, either given by nature or imagined as an imitation of nature? If the lance of an ordinary warrior is ten feet long, what will be the length of Ajax's lance? If I can throw a stone weighing four pounds, Diomedes should be able to shift a boulder. The strides of the gods and the leaps of their horses will be in the same proportion as the imaginary one between gods and men. It's a fourth harmonic string, proportional to three others, with which an animal waits for the resonance which it always makes in itself but which does not always occur in nature. That doesn't matter much to the poet—but it's still the truth. But it's another matter for the philosopher. He must then interrogate nature, which often gives him a phenomenon entirely different from what he had assumed. And then he perceives that the analogy has deceived him.

D'ALEMBERT

Farewell, my friend. Good evening. Have a good night.

DIDEROT

You're teasing, but you'll dream on your pillow about this conversation, and if it doesn't make consistent sense, then too bad for you, because you'll be forced to adopt some even more ridiculous hypotheses.

D'ALEMBERT

You're wrong. I'll go to bed a skeptic, and I'll get up a skeptic.

DIDEROT

A skeptic! Can one really be a skeptic?

D'ALEMBERT

Well, another point? You're not going to maintain that I'm not a sceptic, are you? Who knows that better than me?

DIDEROT

Wait a minute.

D'ALEMBERT

Hurry up. I'm keen to get to sleep.

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DIDEROT

I'll be brief. Do you think there is a single question ever discussed which a man remains both for and against with an equally strict reasonableness.

D'ALEMBERT

No. That would be like the ass of Buridan.⁹

DIDEROT

Then in that case, there's no scepticism, since, except for mathematical questions, which do not admit the least uncertainty, there is a for and an against in all questions. The balance is never equal, and it is impossible that it does not go down on the side which we believe the most probable.

D'ALEMBERT

But in the morning I see probability on my right, and in the afternoon it's on my left.

DIDEROT

That is, you're dogmatically in favour in the morning and dogmatically against in the afternoon.

D'ALEMBERT

And in the evening, when I remember how quick this inconstancy is in my judgments, I believe none of them, neither those of the morning nor of the afternoon.

DIDEROT

That means you don't remember any more the relative persuasiveness of the two opinions between which you oscillated, and this persuasiveness appears too light for you to establish a fixed view, so you take the position of no longer occupying yourself with such problematical subjects, by abandoning the discussion to others and not disputing the question any more.

D'ALEMBERT

That could be.

DIDEROT

But if someone took you to one side and questioned you, as a

⁹ Jean Buridan (c. 1300-c. 1358): a French priest whose name is associated with a paradox: if a rational ass is placed exactly half way between two identical piles of hay, it will starve to death because it has no reason to prefer one pile over the other. In some versions the ass is both hungry and thirsty and the choice is between a bale of hay and a bucket of water.

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friend, and asked you, in conscience, which of these two positions is the one you find less difficult, in good faith would you have any difficulty in answering? Would you behave like Buridan's ass?

D'ALEMBERT

I don't think so.

DIDEROT

Well there you are, my friend. If you think properly about it, you'll find that in everything our true opinion is not one where we have never vacillated but the one to which we most habitually return.

D'ALEMBERT

I believe you're right.

DIDEROT

So do I. Good night, my friend *et memento quia pulvis es, et in pulverem revertis* [and remember that you are dust and will return to dust].

D'ALEMBERT

That's sad.

DIDEROT

And necessary. Give a man, I don't say immortality, but only twice his lifespan, and you'll see what'll happen.

D'ALEMBERT

What do you want to happen? But why should that matter to me? Let what can happen, happen. I want to sleep. Good night.

Denis Diderot
D'ALEMBERT'S DREAM

[Doctor Borden, Mademoiselle de L'Espinasse, D'Alembert, Servant]¹⁰

[The scene is in D'Alembert's bedroom in the morning. D'Alembert is sleeping in a bed with curtains around it. Doctor Borden and Mademoiselle de L'Espinasse are sitting near the bed]

BORDEU

All right, then, is there anything new? Is he ill?

MADemoisELLE DE L'ESPInASSE

I'm afraid so. He's had a very disturbed night.

BORDEU

Has he woken up?

MADemoisELLE DE L'ESPInASSE

Not yet.

BORDEU: *[after going to D'Alembert's bed and feeling his pulse and skin]:*

It won't be anything.

MADemoisELLE DE L'ESPInASSE

You don't think so?

BORDEU

Believe me. The pulse is good . . . a little faint . . . the skin is damp . . . his breathing is easy . . .

MADemoisELLE DE L'ESPInASSE

Is there nothing we can do for him?

BORDEU

Nothing.

MADemoisELLE DE L'ESPInASSE

So much the better. He hates medicines.

¹⁰ Jean Julie de L'Espinasse (1732-1776): a well known patron of artists and intellectuals in Paris; Jean d'Alembert (1717-1783): a celebrated mathematician and *philosophe*, who late in life moved into the home of his close friend Mlle. de L'Espinasse; Théophile de Borden (1722-1776): a successful doctor and close colleague of Diderot.

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BORDEU

So do I. What did he eat for supper?

MADemoiselle de L'ESPINASSE

He didn't want to eat anything. I don't know where he spent the evening, but he came back concerned about something.

BORDEU

He has a slight fever—it won't lead to anything.

MADemoiselle de L'ESPINASSE

As he came back in, he put on his dressing gown and his night cap and threw himself in his armchair, where he dozed off.

BORDEU

Sleep is always beneficial. But it would have been better if he'd been in bed.

MADemoiselle de L'ESPINASSE

He got angry with Antoine for telling him that—we had to pester him for half an hour to make him get to bed.

BORDEU

That happens to me every day, although my health is good.

MADemoiselle de L'ESPINASSE

When he was in bed, instead of lying peacefully the way he usually does, for he sleeps like a child, he began to turn, rolling around and waving his arms. He threw off his blankets and started to talk out loud.

BORDEU

What was he talking about? Was it geometry?

MADemoiselle de L'ESPINASSE

No. It all sounded delirious. At the start it was a lot of nonsense about vibrating strings and sensitive fibres. It all seemed so foolish to me, but since I'd decided not to leave him during the night and not knowing what to do, I went to a small table at the foot of his bed and started to write down everything I could catch of his dream talk.

BORDEU

Clever thinking on your part. Can we see the result?

MADemoiselle de L'ESPINASSE

Of course. But I'll stake my life you'll not understand any of it.

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BORDEU

Perhaps.

MADemoiselle de L'ESPINASSE

Are you ready, doctor?

BORDEU

Yes.

MADemoiselle de L'ESPINASSE

Listen—"A living point . . . No, I'm wrong. Nothing at first, then a living point . . . Another living point attaches itself to this one, and then another—and from these successive conjoinings a single living unity results, for I am certainly a unity. Of that I have no doubt. . . ." (As he was saying this, he was feeling himself all over). "But how is this unity created?" ("My friend," I said to him, "what does that matter to you? Go to sleep." He stopped talking. After a moment of silence, he started up again as if he was talking to someone) . . . "All right, philosopher, I can grasp an aggregate, a tissue of small sensitive beings, but an animal . . . a totality, a unified system, on its own, with an awareness of its own unity? That I don't understand. No, I don't understand it at all. . . ." Doctor, is there something in that you understand?

BORDEU

Yes, it makes excellent sense.

MADemoiselle de L'ESPINASSE

You're really lucky. "Perhaps my difficulties come from a false notion. . . ."

BORDEU

Is this you talking now?

MADemoiselle de L'ESPINASSE: No, it's the dreamer. I'll

keep going . . . He then added, challenging himself, "My friend, D'Alembert, be careful. You're assuming there is only contiguity where there is continuity . . . Yes . . . He is clever enough to tell me that . . . And how is this continuity formed? That will hardly create a problem for him . . . Just as a drop of mercury fuses itself with another drop of mercury, so a sensitive and living molecule fuses itself with a sensitive and living molecule . . . At first there were two drops—after the contact there is only one . . . Before the assimilation there were two molecules; after the assimilation there is only one . . . The sensitivity becomes common to the common mass . . . And, indeed, why not? . . . In my thinking about the length of an animal fibre, I can

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distinguish as many parts as I like, but the fibre will remain a continuous, a unity . . . yes . . . a unity. The contact between two homogeneous molecules, perfectly homogeneous, creates the continuity . . . and it's an example of the most complete union, cohesion, combination, and identity one could imagine . . . Yes, philosopher, if these molecules are elementary and simple . . . but what if they are aggregates, if they are compounds? . . . The combining will still take place no less than before along with the resulting identity and continuity . . . and then the usual action and reaction . . . It's certain that contact between two living molecules is something completely different from the contiguity of two inert masses . . . Let's move on, not bother with that . . . One could perhaps take issue with you, but I'm not worried about that . . . I've never been one to keep on debating the issue. However, let's get back to the point. A wire made of very pure gold—that's one comparison I remember he made to me—a homogeneous network. Between its molecules other molecules interpose themselves and perhaps form another homogeneous network, a tissue of sensitive matter, a contact which absorbs active sensitivity from here and latent sensitivity from there and which passes itself on like a movement, without accounting for the fact, as he has firmly pointed out, that there must be some difference between the contact of two sensitive molecules and the contact of two molecules which have no sensation, and this difference—what can it be? . . . a customary action and reaction . . . and this action and this reaction with a unique character . . . That way everything comes together to produce a sort of unity which exists only in an animal. . . . My goodness, if this is isn't the truth, it's really similar to it." You're laughing, doctor. Do you find any sense in that?

BORDEU

Yes, a lot.

MADemoisELLE DE L'ESPINASSE

So he's not losing his mind?

BORDEU

Not at all.

MADemoisELLE DE L'ESPINASSE

After this preamble, he started to shout, "Mademoiselle de L'Espinasse! Mademoiselle de L'Espinasse!" "What do you want?" "Have you sometimes seen a swarm of bees going out of their hive? . . . The world, or the general mass of matter, is the hive. . . Have you seen them move out to the end of a tree branch to form a long cluster of small winged animals, all hooked to one another by their feet? . . . This cluster is a being,

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an individual, an animal of some sort . . . But these clusters all have to be similar to each other . . . Yes, if he allowed only one homogenous material. . . Have you seen them?" "Yes, I've seen them." "You have seen them?" "Yes, my friend, I tell you I have." "If one of these bees decides somehow to pinch the bee to which it is hanging, what do you think will happen? Tell me." "I have no idea." "Tell me, anyway . . . So you don't know, but the philosopher knows . . . yes, he does. If you ever see him—and you'll either see him or you won't, because that's what he promised—he'll tell you that the second bee will pinch the one next to it, that in the entire cluster there would be as many sensations aroused as there are small animals, that the totality will get aroused, shift itself, change position and shape, that a noise will arise, small cries, and that someone who had never seen a group like that arrange itself would be tempted to assume it was an animal with five or six hundred heads and a thousand or twelve hundred wings. . . ." Well, doctor?

BORDEU

Good. Do you know that this dream is really beautiful? You did well to write it down.

MADemoiselle de L'ESPINASSE

Are you dreaming as well?

BORDEU

So little that I'd almost commit myself to tell you what follows.

MADemoiselle de L'ESPINASSE

I'll challenge you on that.

BORDEU

You challenge me?

MADemoiselle de L'ESPINASSE

Yes.

BORDEU

And if I get it right?

MADemoiselle de L'ESPINASSE

If you get it right I promise you . . . I promise I'll consider you the greatest fool on earth.

BORDEU

Look at your pages and listen to me. "A man who took this cluster for an animal would be wrong." But, Mademoiselle, I assume he went on talking to you. "Do you wish him to judge

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more soundly? Do you wish to transform the cluster of bees into a single unique animal? Soften the feet by which they hold themselves together. Change them from the contiguous condition they are in so that they become continuous. Between this new state of the cluster and the earlier one there is certainly a marked difference. And what might this difference be other than that now it is a totality, a unified animal; whereas, before it was only an assembly of animals? . . . All our organs. . . .”

MADemoiselle de L'ESPINASSE

All our organs!

BORDEU

“For anyone who has practised medicine and made a few observations . . .”

MADemoiselle de L'ESPINASSE

What comes next!

BORDEU

What next? “. . . Are only distinct animals which the law of continuity holds together in a general state of sympathy and unity, a single identity.”

MADemoiselle de L'ESPINASSE

I'm amazed—that's it, and almost word for word. So now I can confirm for all the world that there is no difference between a doctor who's awake and a philosopher who's dreaming.

BORDEU

People suspect that already. It that all?

MADemoiselle de L'ESPINASSE

No, no. You're not there yet. After that nonsense of yours or his, he said to me, “Mademoiselle.” “Yes, my friend.” “Come here . . . closer . . . closer . . . I have something to ask you.” “What is it?” “Take this cluster, the one there—you see it clearly over there—right there. Let's conduct an experiment.” “What?” “Take your scissors—do they cut well?” “Perfectly.” “Move close to the cluster, but gently, very gently, and cut these bees apart. But be careful not to cut them in the middle of their bodies. Cut right at the place where they are joined together by the feet. Don't be afraid of anything—you'll hurt them a little, but you won't kill them . . . Very good. Your actions are as deft as a fairy's. . . Do you see how they fly away, each one in a different direction? They fly off one by one, two by two, three by three. How many of them there are! If you've understood me well . . . have you understood me well?” “Really well.” Now,

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assume . . . assume . . .” My word, doctor, I understood so little of what I was writing down. He spoke in such a low voice, and this section of my paper is so scribbled I can’t read it.

BORDEU

I’ll make up for that if you like.

MADemoisELLE DE L’ESPINASSE

If you can.

BORDEU

Nothing is easier. Assume that these bees are small, so small that the crude cutting edge of your scissors always misses their organic structures. You keep up your cutting as far as you like without killing one of them, and this totality, made up of imperceptible bees, will be a true polyp which you only destroy by crushing. The difference between the group of bees formed continuously and the group of bees formed contiguously is precisely the difference between normal animals—like us, fish, worms, and snakes—and animal polyps. Moreover, if this whole theory undergoes a few modifications . . . [*Mademoiselle de L’Espinasse abruptly jumps up and goes to pull on the bell rope*]. Gently, gently, Mademoiselle, you’ll wake him up, and he needs his rest.

MADemoisELLE DE L’ESPINASSE

I wasn’t thinking of that—my mind is spinning so. [*To the servant who enters*]. Which of you went to the doctor’s house?

SERVANT

It was me, Mademoiselle.

MADemoisELLE DE L’ESPINASSE

A long time ago?

SERVANT

I came back less than an hour ago.

MADemoisELLE DE L’ESPINASSE

Did you take anything there?

SERVANT

Nothing.

MADemoisELLE DE L’ESPINASSE

No piece of paper?

SERVANT

No, none.

D'ALEMBERT'S DREAM

MADemoiselle de L'ESPINASSE

All right, then. You can go . . . I'm astonished. You see, doctor, I suspected one of them had told you about my scribblings.

BORDEU

I assure you there was nothing like that.

MADemoiselle de L'ESPINASSE

Doctor, now that I realize your talent, you'll be a great help to me in my social life. His dreaming did not stop at that point.

BORDEU

So much the better.

MADemoiselle de L'ESPINASSE

So you don't see anything in this to get alarmed about?

BORDEU

Not in the least.

MADemoiselle de L'ESPINASSE

He went on, "Well then, philosopher, so you have an idea of polyps of all kinds, even human polyps? . . . But nature doesn't show us any such things."

BORDEU

He had no knowledge of those two girls joined together at the head, shoulders, back, buttocks, and thighs, who lived fused together like this until the age of twenty-two and who died within a few minutes of each other. He went on to say . . . ?

MADemoiselle de L'ESPINASSE

Ravings which are only heard in the lunatic asylum. He said, "That's over or it will happen. And then who knows the state of things in other planets?"

BORDEU

Perhaps we don't have to go that far.

MADemoiselle de L'ESPINASSE

"In Jupiter or Saturn, human polyps! The males resolve themselves into males, females into females—that's an amusing thought . . ." (At that point he began to burst out laughing so hard I was frightened) "Man splitting himself up into an infinity of atomized men which we could keep between sheets of paper like eggs from insects which spin their cocoons, remain for a certain period in the chrysalis state, pierce through their cocoons, and escape as butterflies—a human society formed

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and an entire region populated by the fragments of a single individual—all that is very pleasant to imagine. . . .” (Then the bursts of laughter started again) “If there’s a place where the human being divides itself into an infinity of human animalcules, people there should be less reluctant to die. It’s so easy to make up for the loss of a person that death should cause little regret.”

BORDEU

This extravagant assumption is almost the real history of all species of animals—those presently existing and those still to come. If man does not divide himself into an infinity of human beings, at least he does divide himself up into an infinity of animalcules, whose changes and future and final organic structure is impossible to predict. Who knows if that is not the breeding ground for a second generation of beings separated from this one by an incomprehensible interval of centuries and successive developments?

MADemoiselle DE L’ESPINASSE

What are you muttering about in such a low voice, doctor?

BORDEU

Nothing. Nothing at all. It was my turn to dream. Mademoiselle, continue reading.

MADemoiselle DE L’ESPINASSE

“All things well considered, however, I do prefer our method of reproducing,” he added. . . . “Philosopher, you who know about what’s going on out there or elsewhere, tell me about the splitting up of the different parts—doesn’t that produce men of different characters? The brain, heart, chest, feet, hands, testicles . . . O how that simplifies morality! . . . A man born, a woman derived from” (Doctor, will you allow me to overlook this part?) “A warm room, lined with small container cups, and on each of these cups a label: warriors, magistrates, philosophers, poets, cup of courtiers, cup of prostitutes, cup of kings.”

BORDEU

That’s all very cheerful foolery. That’s what it means to dream—and a vision which leads me to some rather peculiar phenomena.

MADemoiselle DE L’ESPINASSE

Then he began to babble something or other about seeds, scraps of flesh minced and placed in water, different races of animals which he saw in succession as they were born and passed away. With his right hand he imitated the tube of a microscope and, with his left, I think, the opening of a vase. He looked into the

vase through the tube and said, "Voltaire may make as much fun as he likes about it, but the Eel-monger is right—I believe my eyes, I see them—how many of them there are! How they come and go! How they wriggle around! . . ."¹¹ The vase where he was looking at so many momentary generations he then compared to the universe. He saw in a drop of water the history of the world. This idea appeared grand to him. He found it entirely compatible with good philosophic practice, which studies large bodies by examining small ones. He said, "In Needham's drop of water, everything takes place and goes away in the blink of an eye. In the world, the same phenomenon lasts a little longer, but what is our length of time compared to an eternity of time? Less than the drop which I took up on the point of a needle compared to the limitless space which surrounds me. An indefinite succession of animalcules in the fermenting atom, a similar indefinite succession of animalcules in the other atom which we call the Earth. Who knows the races of animals which came before us? Who knows the races of animals which will come after ours? Everything changes, everything passes away. Only the totality remains. The world begins and ends without ceasing. At every instant it is at its beginning and at its end. It has never been anything else and never will be anything else. In this immense ocean of matter, no single molecule resembles any other, and no single molecule resembles itself for more than a moment: *Rerum novus nascitur ordo* [*a new order of things is born*]*—there's its eternal slogan.*" Then he sighed and added: "O the vanity of our thoughts! O the poverty in glory and in our works! O the wretched smallness of our vision! There's nothing substantial except drinking, eating, living, loving, and sleeping . . . Mademoiselle de L'Espinasse, where are you?" "I'm here" Then his face became flushed. I wanted to feel his pulse, but I didn't know where he had hidden his hand. It looked as though he was going through a convulsion. His mouth was half open, and his breath was forced. He gave a deep sigh, and then a fainter sigh, and then another deeper one. He turned his head on his pillow and went to sleep. I looked at him attentively, and I was very moved without understanding why. My heart was beating—but it wasn't fear. At the end of a few moments I saw a slight smile cross his lips. He spoke in a low voice, "In a planet where human beings reproduced themselves the way fish do, where the spawn of a

¹¹ The Eel-monger is a reference to John Needham (1713-1781) an English priest and biologist whose experiments seemed to confirm that life could arise spontaneously from non-living matter. Spontaneous generation of living things from inert matter was widely accepted in the eighteenth century. The phrase eel-monger (in French *anguillard*) was used by Voltaire as a contemptuous term for Needham.

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man fell upon the spawn of a woman . . . I'd have fewer regrets there . . . We mustn't lose anything of what could prove useful. Mademoiselle, if that stuff could be collected, enclosed in a flask, and sent at daybreak to Needham. . . ." Now, Doctor, don't you call all that madness?

BORDEU

In your company I certainly would.

MADemoiselle DE L'ESPINASSE

In my company, far away from me—it's all one thing, and you don't know what you are talking about. I'd hoped that the rest of the night would be peaceful.

BORDEU

That's what usually happens.

MADemoiselle DE L'ESPINASSE

Not this time. At around two o'clock in the morning, he came back to his drop of water—he called it a *mi* . . . a *micro* . . .

BORDEU

. . . a microcosm.

MADemoiselle DE L'ESPINASSE

That's the word he used. He was admiring the wisdom of the ancient philosophers. He was saying or putting words into the mouth of his philosopher—I don't know which of the two, "When Epicurus claimed that the earth contained the germs of everything and that the animal species was a product of fermentation, if he had proposed to show a picture in miniature of what was created on a grand scale at the beginning of time, what could one have said in reply?¹² . . . And you have this very image right before your eyes, but it's not telling you anything . . . Who knows if the process of fermentation and what it produces have run their course? Who knows what point we're at in the sequence of these animal generations? Who knows if this deformed biped, only four feet high, which is still called a human being in the regions of the pole and which would quickly lose this name if it grew a little more deformed, is not the image of a species which is passing away? Who knows if things are not the same with all animal species? Who knows if everything isn't tending to reduce itself to a large, inert, and immobile sediment?

¹²Epicurus (341 BC-270 BC) was an ancient Greek philosopher who taught a thoroughly materialistic philosophy. His ideas were passed on, most notably, by the poem *On the Nature of Things* by Lucretius, a very popular work in the eighteenth century.

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Who knows how long this inertia will last? Who knows what new race could result once more from such a huge heap of sensitive and living points? Why not a single animal? What was the elephant at its origin? Perhaps it was the huge animal as it appears to us, perhaps an atom, for both options are equally possible. They only depend upon the movement and various properties of matter . . . The elephant, this enormous structurally organized mass, the sudden product of fermentation! Why not? The size relationship between this large quadruped and its original womb is less than that between the mite and the particle of flour which produced it. But the mite is only a mite . . . That is, its diminutive size which conceals from you its organic structure robs it of its wonder. . . . The amazing thing is life, sensitivity—and this is no longer something amazing . . . Once I have seen inert matter passing into the sensitive state, nothing else should astonish me . . . What a comparison between a small number of fermenting elements set in the palm of my hand and that immense reservoir of different elements scattered in the bowels of the earth, on its surface, in the bosom of the seas, in the expanses of air! . . . However, since the same causes are at work, why have the effects ceased? Why do we not see the bull piercing the earth with his horn any more, his feet pushing against the soil, as he makes an effort to free his heavy body from it? . . . Let the present race of existing animals pass away, and let the large inert sediment do its work for a few million centuries. Perhaps, in order to renew species, it requires ten times longer than the period assigned for their duration. Be careful. Don't be in a rush to make judgments about the great work of nature. You have two grand phenomena—the passage of an inert state into a sensitive state and spontaneous generations—and that's enough for you. Draw justified conclusions from them and in an order where there is no large or small, no absolutely durable or temporary. Watch out for the logical fallacy of the ephemeral . . .” Doctor, what is the logical fallacy of the ephemeral?

BORDEU

It occurs when a transitory being believes in the immortality of things.

MADemoiselle de L'ESPINASSE:

Like Fontenelle's rose which said that in the memory of a rose no one had ever seen a gardener die?¹³

¹³Bernard de Fontenelle (1657-1757) was a popular and successful French writer on a wide range of subjects, including natural philosophy.

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BORDEU

Precisely—that's both deft and profound.

MADemoiselle DE L'ESPINASSE

Why don't your philosophers express themselves with the grace of Fontenelle? Then we'd understand them.

BORDEU

To be frank, I don't know if such a frivolous tone is appropriate for serious subjects.

MADemoiselle DE L'ESPINASSE

What do you call a serious subject?

BORDEU

Well, sensibility in general, the formation of a sentient being, its unity, the origin of animals, how long animal life lasts, and all questions related to these matters.

MADemoiselle DE L'ESPINASSE

Well, I call all that so much nonsense, which I'll admit people dream about when they're asleep but which a sensible man never concerns himself with when he's awake.

BORDEU

Please tell me why you think that.

MADemoiselle DE L'ESPINASSE

Because some of them are so obvious it's useless to seek out the reason, and others are so obscure that there's nothing to see in them, and all are perfectly useless.

BORDEU

Mademoiselle, do you believe that it makes no difference whether you deny or admit there's a supreme intelligence?

MADemoiselle DE L'ESPINASSE

No.

BORDEU

Do you think one can adopt a position on the supreme intelligence without knowing what to believe about the eternal quality of matter and its properties, the difference between the two substances, mind and matter, the nature of man, and the development of animals?¹⁴

¹⁴I have once again added the phrase "mind and matter" to clarify the meaning of "the two substances."

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MADemoiselle de l'EsPInasse

No.

BORDEU

So these questions are not as pointless as you said.

MADemoiselle de l'EsPInasse

But how can they be important to me if I don't know how to clarify them?

BORDEU

And how will you know that if you don't examine them? But could I ask you which ones you find so obvious that examining them seems superfluous to you?

MADemoiselle de l'EsPInasse

Well, for example, the question of my unity, of my "me." My goodness, it seems to me that so much verbiage is not necessary to know that I am myself, that I've always been me, and that I'll never be someone else.

BORDEU

No doubt the fact is obvious, but the reason for the fact is not at all obvious, especially in the hypothesis of those who only allow one substance and who explain the formation of man or animals in general by the successive accumulation of several sensitive molecules. Each sensitive molecule had its identity (its "me") before the accumulation, but how did it lose that, and how, from all these lost identities, does one end up with the consciousness of a totality?

MADemoiselle de l'EsPInasse

It seems to me that contact alone is sufficient. Here's a experiment which I've done a hundred times . . . but wait a moment . . . I have to go to see what's happening behind these curtains . . . he's sleeping . . . When I put my hand on my thigh, at first I clearly sense that my hand is not my thigh, but after a certain length of time, when the heat is the same in both of them, I don't make that distinction any more—the limits of the two parts get mixed up, and they are as one.

BORDEU

Yes, until someone pricks one or the other. Then the distinction returns. So there is something in you which knows whether it's your hand or your thigh which has been pricked, and this something, it's not your foot, not even your pricked hand. It's the hand which hurts, but it's something else that knows it, something which does not itself suffer from the pain.

D'ALEMBERT'S DREAM

MADemoisELLE DE L'ESPINASSE:

But I think that something is my head.

BORDEU

It is your entire head?

MADemoisELLE DE L'ESPINASSE

No. But look, Doctor, I'm going to explain myself with a comparison. Almost all the reasoning of women and poets consists of comparisons. Imagine a spider . . .

D'ALEMBERT

What's going on there? . . . Is that you, Mademoiselle de L'Espinasse?

MADemoisELLE DE L'ESPINASSE

Shhh . . . keep quiet. [*Mademoiselle de L'Espinasse and the doctor remain silent for some time. Then Mademoiselle de L'Espinasse speaks in a low voice*]. I think he's gone back to sleep.

BORDEU

No—it seems to me I hear something.

MADemoisELLE DE L'ESPINASSE

You're right. Has he resumed his dreaming?

BORDEU

Let's listen.

D'ALEMBERT

Why am I the way I am? That's because it was necessary for me to be like this . . . Here, yes, but somewhere else? At the pole? Below the equator? On Saturn? . . . If a distance of a few thousand leagues changes my species, what would a distance of a few thousand earth diameters do? . . . And if everything is a universal flux, as the panorama of the universe demonstrates to me everywhere, what would the changes in a time span of a few million centuries produce here and elsewhere? . . . Who knows what a thinking, feeling being is on Saturn? . . . But is there feeling and thought on Saturn? . . . Why not? . . . Would the sentient and thinking being on Saturn have more senses than I do? . . . If that's so, ah, how unfortunate for the Saturnian! . . . The more senses, the more needs.

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BORDEU

He's right. The organs produce the needs and, conversely, the needs produce the organs.¹⁵

MADemoiselle de L'ESPINASSE

Doctor, are you also delirious?

BORDEU

Why not? I've seen two stumps become over time two arms.

MADemoiselle de L'ESPINASSE

You're not telling the truth.

BORDEU

No I'm not, but when the two arms were missing I've seen the two shoulder blades grow longer, move in a pincer motion, and grow into two stumps.

MADemoiselle de L'ESPINASSE

That's crazy!

BORDEU

It's a fact. Imagine a long sequence of generations of people with no arms. Now assume a continuous effort, and you'll see the two sides of this pincer grow longer, and get longer and longer, cross over one another at the back, return to the front, and perhaps develop digits at their extremities, thus making arms and hands once again. The original structure alters itself or perfects itself according to necessity or habitual functions. We walk and work so little, and we think so much, that I wouldn't deny the possibility that man might finish up being nothing but a head.

MADemoiselle de L'ESPINASSE

A head! Just a head! That's not much. I was hoping that with unrestrained love-making . . . You're putting all sorts of ridiculous ideas in my head.

BORDEU

Quiet!

¹⁵It was widely believed in the eighteenth century that changes in organic structures could take place during the lifetime of the animal by the way the animal lived, i.e., characteristics could be acquired by habits or even by will power and then passed on. The giraffe, for example, by constantly reaching for food higher up in the trees would lengthen its neck and then pass on that longer neck to its offspring.

D'ALEMBERT

So I am the way I am because I had to be that way. Change the whole and you necessarily change me. But the totality is changing constantly . . . Man is only a common effect; a monster is only a rare effect. Both of them are equally natural, equally necessary, equally part of the universal general order . . . Is there anything astonishing in that? . . . All beings circulate through each other—thus all the species . . . everything is in a perpetual flux . . . Every animal is more or less a human being, every mineral is more or less a plant, every plant is more or less an animal. There is nothing fixed in nature . . . The ribbon of Father Castel . . . Yes, Father Castel, it's your ribbon and nothing else.¹⁶ Everything is more or less something or other, more or less earth, more or less water, more or less air, more or less fire, more or less of one kingdom or another . . . so there is no essence of any particular being . . . No, there's no doubt, since there is no quality which any being does not share in . . . and because it's the greater or smaller ratio of this quality which has made us attribute it to one being to the exclusion of another . . . And you talk about individuals, you poor philosophers! Forget about your individuals. Answer me this: is there an atom in nature which is exactly similar to another atom? No . . . Don't you agree that everything in nature is linked and it's impossible that there's a gap in nature's chain? Then what do you want to say with your individuals? There are no individuals, no, there are none . . . There is only one great individual—that's the totality. In this totality, as in a machine, in some animal or other, there is a part which you'll call this or that, but when you give the name "individual" to this part of the totality, it's a conceptual error, just as if, in a bird, you gave the name "individual" to a wing, to a feather in the wing . . . And you speak of essences, your poor philosophers! Forget about your essences. Look at the general mass, or if your imagination is too narrow to embrace it all, consider your first origin and your final end . . . O Archytas, you who measured the globe, what are you now?¹⁷ A few cinders . . . What is a being? . . . The sum of a certain number of tendencies . . . Can I be anything other than a tendency? . . . No, I'm moving towards an end . . . And what about the species? . . . Species are only tendencies towards a

¹⁶Louis Bertrand Castel (1688-1757) was a Jesuit priest who proposed that musical sounds could be represented by colours (since, according to him, they were both the product of vibrations, a scientific theory that opposed Newton's theory of colour) and that a harpsichord could be devised to represent musical compositions with colours. Serious attempts were made to construct a machine to demonstrate Castel's theory, which enjoyed for some time a great deal of support. D'Alembert's point here is that all things in nature merge into other things.

¹⁷Archytas (428-347 BC) was an ancient Greek philosopher, mathematician, and general.

D'ALEMBERT'S DREAM

common end appropriate to them . . . And life? . . . Life, a series of actions and reactions . . . When living, I act and react as a mass . . . when dead, I act and react as different molecules . . . So I don't die? . . . No, undoubtedly I don't die in that sense, neither I nor anything that is . . . To be born, live, and pass away—that's changing forms . . . And what's important about one form or another? Each form has the happiness and unhappiness appropriate to it. From the elephant all the way to the aphid . . . from the aphid all the way to the sensitive and living molecule, the origin of everything, there's no point in all nature which does not undergo pain or pleasure.

MADemoiselle de L'ESPINASSE

He's stopped talking.

BORDEU

Yes. He made a really fine speech. Now that's lofty philosophy. At this point it's a theoretical system, but I believe that the more human knowledge progresses, the more it will be confirmed.

MADemoiselle de L'ESPINASSE

And what about us? Where were we?

BORDEU

To tell you the truth, I don't remember any more. He reminded me of so many things while I was listening to him.

MADemoiselle de L'ESPINASSE

Wait . . . just a minute . . . I was mentioning my spider.

BORDEU

Yes, yes.

MADemoiselle de L'ESPINASSE

Doctor, come closer. Imagine a spider at the centre of its web. Shake a strand. You'll see the animal rush up on the alert. All right then. What if the strands which the insect pulls from its intestines and pulls back when it wishes were a sensitive part of itself?

BORDEU

I understand. You are imagining in yourself some part in a corner of your head—for example, in that part we call the meninges—one or several points where all the sensations aroused along the length of strands are brought.¹⁸

¹⁸The meninges: the membranes that surround the central nervous system (the brain and spinal chord).

D'ALEMBERT'S DREAM

MADemoiselle de L'ESPINASSE

That's it.

BORDEU

Your idea is as good as one can make it, but don't you see that it's almost the same thing as a particular cluster of bees?

MADemoiselle de L'ESPINASSE

Ah yes, that's true. I've been composing prose without realizing it.

BORDEU

And very good prose, as you're going to see. Anyone who understands a human being only by the form which he shows us at birth does not have the least idea. His head, feet, hands, all his limbs, all his viscera, all his organs, his nose, eyes, ears, heart, lungs, intestines, muscles, bones, nerves, membranes, properly described, are only the basic developments of a network which is formed, grows, extends itself, and throws out a multitude of imperceptible threads.

MADemoiselle de L'ESPINASSE

That's my web, and the starting point of all these threads is my spider.

BORDEU

Precisely.

MADemoiselle de L'ESPINASSE

Where are the strands? Where's the spider located?

BORDEU

The strands are everywhere. There is no part on the surface of your body where they don't end up. And the spider is lodged in a part of your head—the one I mentioned to you—the meninges, which we can hardly touch without knocking the entire machine unconscious.

MADemoiselle de L'ESPINASSE

But if an atom sets one of the strands in the spider's web vibrating, then the spider is alarmed and disturbed. It flees or runs up. At the centre it is informed about everything which goes on in any point of the immense dwelling it has woven. Why don't I know what's going on in mine or in the world, since I am a pack of sensitive points which all impinge on me and since I impinge on everything?

D'ALEMBERT'S DREAM

BORDEU:

It's because the impressions grow weaker in proportion to the distance from where they originate.

MADemoiselle de L'ESPINASSE

If we strike the lightest blow at the end of a long girder and if I place my ear on the other end, I hear the blow. If one end of the girder was touching the earth and the other end was in Sirius, the same effect would be produced. If everything is linked, contiguous—that is, if the girder really existed—why do I not hear what goes on in the immense space which surrounds me, especially if I really open my ears?

BORDEU

And who has told you that you do not hear it more or less? But the distance is so great, the impression so faint, the passage so confused. You are surrounded and deafened by such violent and different sounds. And also between you and Saturn there are only contiguous bodies; whereas, there would need to be continuity.

MADemoiselle de L'ESPINASSE

That's a great pity.

BORDEU

That's true, because you would be God. Through your identity with all natural beings you would know everything going on. Through your memory you'd know everything that has been.

MADemoiselle de L'ESPINASSE

And what will be?

BORDEU

You'd form plausible conjectures about the future, but subject to error. It's precisely as if you were seeking to guess what's going to happen inside you or at the end of your foot or your hand.

MADemoiselle de L'ESPINASSE

And who told you that this world doesn't also have its meninges or that there isn't a huge or a small spider living in some corner of space with its strands extending out to everything.

BORDEU

No one, still less if it has ever existed or will exist in future.

MADemoiselle de L'ESPINASSE

How could a God of that sort . . .

D'ALEMBERT'S DREAM

BORDEU

The only sort one can conceive of . . .

MADemoiselle de l'ESPINASSE

. . . how could He have existed or come into being and pass away?

BORDEU

How indeed? But since He would be made up of matter in the universe, a portion of the universe, subject to change, He'd grow old and die.

MADemoiselle de l'ESPINASSE

But hold on—another fanciful idea has occurred to me.

BORDEU

I'll spare you describing it. I know what it is.

MADemoiselle de l'ESPINASSE

We'll see. What is it?

BORDEU

You see intelligence unified with very energetic portions of matter and the possibility of all sorts of imaginable wonders. Others have thought the way you do.

MADemoiselle de l'ESPINASSE

You have guessed my thoughts correctly, but I don't respect you any more for that. You must be amazingly fond of foolishness.

BORDEU

I agree. But what's frightening about that idea? There would be an epidemic of good and evil geniuses, the most constant laws of nature would be interrupted by natural agents, it would make our general physics more difficult to grasp, but there would be no miracles.

MADemoiselle de l'ESPINASSE

In truth, we should be very cautious about what we affirm and what we deny.

BORDEU

Come now, anyone who told you about a phenomenon like that would sound like a great liar. But let's set aside these imaginary beings, including your spider with its infinite networks. Let's go back to your network and its formation.

D'ALEMBERT'S DREAM

MADemoisELLE DE L'ESPINASSE

'That's fine with me.

D'ALEMBERT

Mademoiselle, you're there with someone. Who's talking out there with you?

MADemoisELLE DE L'ESPINASSE

It's the doctor.

D'ALEMBERT

Good morning, doctor. What are you doing here so early in the morning?

BORDEU

You'll find out. Go to sleep.

D'ALEMBERT

On my word, I need to. I don't think I've spent a night as disturbing as this past one. Don't leave before I get up.

BORDEU

No, I won't. Mademoiselle, I'll wager that since you believed that at twelve years old you were a woman half the size you are now, and at four years old a woman half as small again, at the foetal stage a tiny woman, and in the sex cells of your mother a very small woman, you thought that you've always been a woman with the form you have now, so that the successive stages of growth you've been through are the only things that have made the entire difference between you at your origin and you as you are now.¹⁹

MADemoisELLE DE L'ESPINASSE

Yes, that's right.

BORDEU

However, nothing is more false than this idea. To begin with, you were nothing. At the start you were an imperceptible point formed of the smallest molecules, scattered in the blood and lymph of your father and mother. This point became a slender thread, then a bundle of threads. Up to that point, there wasn't the least trace of this pleasing shape you have now. Your eyes, these beautiful eyes, didn't look any more like eyes than the tip of an anemone's claw looks like an anemone. Each strand of

¹⁹Bordeu is criticizing here the preformation hypothesis, which maintains that the sex cells contain seeds of the future person in miniature, so that human development consists merely of growing larger.

D'ALEMBERT'S DREAM

this bundle of threads was changed merely by nutrition and its structure into a particular organ, with the exception those places where strands of the bundle change completely and give rise to an organ. The bundle is purely a system of sensation. If it remained in this form, it would be susceptible to all impressions connected with mere sensibility, like cold, heat, softness, roughness. These successive impressions, different from each other and of varying intensities, perhaps could produce memory in it, awareness of itself, and a very limited reasoning. But this pure and simple sensibility, this sense of touch, diversifies itself in the organs developed from each of these strands of the bundle. One strand forms an ear and gives rise to a type of touching we call noise or sound. Another forms the palate, giving rise to second type of touching we call taste. A third forms the nose and the nasal lining, giving rise to a third type of touching we call smell. A fourth forms an eye and gives rise to a fourth type of touching we call colour.

MADemoiselle de L'ESPINASSE

But if I've understood you correctly, those who deny the possibility of a sixth sense, a true hermaphrodite, are foolish. Who told them that nature could not form a bundle with a unique thread which would give rise to an organ unknown to us?

BORDEU

Or with two strands characteristic of the two sexes? You're right. It's a pleasure to talk with you. You not only grasp what people say to you, but you also draw conclusions from that with a justness which astonishes me.

MADemoiselle de L'ESPINASSE

Doctor, you're encouraging me.

BORDEU

No, on my word—I'm telling you what I think.

MADemoiselle de L'ESPINASSE

I see well enough the use of some of these strands of the bundle, but what do the others become?

BORDEU

Do you think someone other than yourself has wondered about this question?

MADemoiselle de L'ESPINASSE

Certainly.

D'ALEMBERT'S DREAM

BORDEU

Well, you aren't conceited. The rest of the threads go on to form as many other types of touching as there is variety between the organs and the parts of the body.

MADemoiselle de l'ESPINASSE

What do we call them? I've never heard people talk about that.

BORDEU

They don't have a name.

MADemoiselle de l'ESPINASSE

Why is that?

BORDEU

Because there are not as many differences among the sensations they arouse as there are between the sensations aroused by the other organs.

MADemoiselle de l'ESPINASSE

In all seriousness, do you think the foot, hand, thighs, stomach, chest, lungs and heart have their own characteristic sensations?

BORDEU

Yes, I do. If I dared, I'd ask you if among these sensations which we have no name for . . .

MADemoiselle de l'ESPINASSE

I see what you mean. No. That is the only one of its type, and that's a pity. But what reason do you have for this multiplicity of sensations—more painful than pleasant—which it pleases you to bestow on us.

BORDEU

What reason? Well, because we perceive the majority of them. If this infinite diversity of touching did not exist, we'd know that we were experiencing pleasure or pain, but we wouldn't know what to connect them with. We'd have to have recourse to our vision. That wouldn't be a matter of sensation any more, but a matter of experience and observation.

MADemoiselle de l'ESPINASSE

So if I said I have a pain in my finger and if someone asked me why I know the pain is in my finger, I'd have to reply not that I felt it, but that I felt the pain and I saw that my finger was hurt.

BORDEU

That's it. Come let me give you a kiss.

D'ALEMBERT'S DREAM

MADemoiselle de l'EsPInasse

With pleasure.

D'ALEMBERT

Doctor, you are kissing mademoiselle—good for you.

BORDEU

I've given the matter a great deal of thought, and it seems to me that the direction and the location of the stimulus would not be sufficient to permit so sudden a judgment of whatever it is at the heart of the bundle.

MADemoiselle de l'EsPInasse

I have no idea about that.

BORDEU

Your doubt pleases me. It is so common to assume that natural qualities are acquired habits almost as old as we human beings.

MADemoiselle de l'EsPInasse

And vice versa.

BORDEU

Whatever it is, you see that with a question where it's a matter of the first formation of an animal, it's too late to concentrate one's focus and thoughts on the completely developed animal—we have to go back to its first rudiments. For that you have to strip off your present organic structure and get back to a moment when you were merely a soft substance made up of vermicular filaments, without a shape, more analogous to a bulb or a plant root than to an animal.

MADemoiselle de l'EsPInasse

If our custom was to go down the street totally naked, I would not be the first or the last to conform. So deal with me as you wish, provided I learn something. You told me that each strand of the bundle formed a particular organ. What proof do you have of this?

BORDEU

In your mind do what nature sometimes does—cut through one of the strands of the bundle, for example, the strand which forms the eyes. What do you think happens?

MADemoiselle de l'EsPInasse:

Perhaps the animal will have no eyes.

D'ALEMBERT'S DREAM

BORDEU

Or it will have only a single one placed in the middle of the forehead.

MADemoiselle de L'ESPINASSE

It will be a cyclops.

BORDEU

A cyclops.

MADemoiselle de L'ESPINASSE

So it could be that the cyclops was not a creature out of fables.

BORDEU

Probably not—and I'll show you one if you like.²⁰

MADemoiselle de L'ESPINASSE

Who knows the cause of this change?

BORDEU

The man who dissected this monstrosity and found it had only one optic thread. Now, in your mind do what nature sometimes does. Remove another strand of the bundle, the thread that forms the nose, and the animal will be without a nose. Remove the thread which should form the ear, and the animal will be without ears or will only have one, and the anatomist in his dissection won't find either the olfactory threads or the auditory threads or will find only one of the latter. If you continue removing strands, the animal will lack a head, feet, hands—its lifespan will be short, but it will have lived.

MADemoiselle de L'ESPINASSE

And are there any examples of this?

BORDEU

Certainly. And that's not all. If you double some of these strands in the bundle, the animal will have two heads, four eyes, four ears, three testicles, three feet, four arms, and six fingers on each hand. Mix up the threads of the bundle, and the organs will be displaced: the head will be situated in the middle of the chest,

²⁰A Cyclops: a mythological Greek monster with one eye in the middle of its forehead. In a Supplement to Volume 4 of his famous scientific work *L'Histoire naturelle* (1777), the Comte de Buffon published an engraving of a child with one eye in the middle of his head (a congenital condition known as *cyclopia* or *cyclocephaly*). The picture was derived from the work of Marie Marguerite Bihéron (1719-1795) who observed such a child born in 1766 and had made a model of it. Diderot had studied anatomy with her and had been her neighbour (R. Loyalty Cru, *Diderot as a Disciple of English Thought*, Columbia UP, 1913).

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the lungs will be on the left, the heart on the right. Stick two threads together, and the organs will be mixed together—the arms will be stuck on the body, the thighs, limbs, and feet will be fused together, and you'll have every kind of monstrosity you can imagine.

MADemoiselle de l'Espinasse

But it seems to me that a machine as complex as an animal, a machine which develops from a single point, in an agitated fluid, perhaps in two fluids mixed together randomly—for one hardly knows what one is doing at such times—a machine which develops toward perfection through an infinity of successive developments and whose regular or irregular development depends upon a bundle of thin, delicate, flexible wires, a kind of tangle where the least thread cannot be broken, worn out, displaced, or missing without harmful consequences for the totality—such a machine would get all tied up and confused during its development even more often than the silks on my spinner.

BORDEU

Well, it does suffer much more than we think. There's not enough dissection done, and our ideas about its development are very far from the truth.

MADemoiselle de l'Espinasse

Do we have any noteworthy examples of these original deformities, other than hunchbacks and cripples, in whom we could attribute the misshapen form to some hereditary defect?

BORDEU: There are numberless examples. Very recently a man died of pneumonia in Charité de Paris infirmary. He was a twenty-five-year-old carpenter born at Troyes, called Jean-Baptiste Macé. The inner organs of his chest and abdomen were reversed—his heart was on the right, just as it is on the left in you; his liver was on the left, his stomach, spleen, and pancreas on the right hypochondria; the portal vein to the liver on the left side (corresponding to the position it has when it goes to the liver on the right), the same transposition along the length of the intestinal tract; the kidneys leaning against each other by the lumbar vertebrae, making the shape of a horseshoe. And with all that people go on talking to us about final causes!²¹

²¹The final cause is "something that is the end or purpose of a process" (Merriam-Webster). A great deal of traditional natural science focused on describing final causes (i.e., the reasons things existed) and these descriptions were usually framed in clear moral language linked to a religious understanding of creation. Diderot and the new natural scientists rejected this emphasis and replaced it with an emphasis

D'ALEMBERT'S DREAM

MADemoiselle de L'ESPINASSE

'That's remarkable.

BORDEU

Now, if Jean-Baptiste Macé had been married and had had children . . .

MADemoiselle de L'ESPINASSE

Yes, doctor, what about his children?

BORDEU

They'd have had the usual shape, but because these irregularities make jumps, some child of their offspring, after about a hundred years, would return back to the strange organic arrangements of his ancestor.

MADemoiselle de L'ESPINASSE

Where do these jumps come from?

BORDEU

Who knows? To make a child requires two people, as you know. Perhaps one of these agents fixes the defect in the other and the defective network is not reborn until the moment when the descendant of the family with the monstrosity predominates and determines the formation of the network. The bundle of threads is the basis for the first and original difference in all animal species. The varieties in the bundle for a species create all the monstrous varieties of this species . . . [*After a long silence, Mademoiselle de L'Espinasse emerges from her reverie and draws the doctor out of his with the following comments*] . . . I've just had a ridiculous idea.

BORDEU:

What's that?

MADemoiselle de L'ESPINASSE:

Perhaps man is only a monstrous variety of woman or a woman a monstrous variety of man.

BORDEU

This idea would have come to you a lot sooner if you had known that a woman has all the parts of a man and that the only difference is between a pouch which hangs outside and a pouch that is tucked away inside, that a female foetus looks so like a male foetus that one can make a mistake about them, that the

on efficient causes (i.e., the mechanical natural processes that lead to certain effects).

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part which gives rise to the error grows smaller in the female foetus as the interior pouch enlarges, but it never diminishes to the point of losing its original shape, and it keeps this shape in miniature, undergoes the same movements, and is also the source of feelings of sexual pleasure. It has its glans and prepuce, and we can see at its tip a point which seems to have been the opening to a urinary canal which is closed off. And in a man, from the anus up to the scrotum, there's a space called the perineum, and from the scrotum to the tip of the penis, a seam which seems to be a repeat of a sealed up vulva. Also women who have an excessively large clitoris have beards, and eunuchs do not, but their thighs build up, their hips widen, and their knees grow rounder. As they lose the organic structure characteristic of one sex, they seem to revert to the arrangements characteristic of the other. Among Arabs, those who have been castrated by constant horse riding lose their beards, acquire a high-pitched voice, dress in women's clothes, and sit among them in wagons. They crouch down to urinate and take on the customs and habits of women. . . . But we've come a long way from our subject. Let's get back to our bundle of animated and living filaments.

D'ALEMBERT

I'm believe you've been talking about some dirty things with Mademoiselle de L'Espinasse.

BORDEU

When one talks science, it's necessary to use technical language.

D'ALEMBERT

You're right. Then the words lose the train of associated ideas which makes them improper. Continue, doctor. So you were saying to Mademoiselle that the womb is nothing other than a scrotum tucked back from the outside to the inside, a movement in which the testicles have been thrown out of the pouch which enclosed them and placed on the right and the left in the body cavity, that the clitoris is a miniature male member, that this virile member in a woman gets increasing small to the extent that the womb or the reversed scrotum enlarges, and that . . .

MADemoiselle de L'ESPINASSE

Yes, yes. Now stay quiet, and don't interfere in our business.

BORDEU

You see, Mademoiselle, that with the question of our sensations in general, which are all only various kinds of touching, we have

D'ALEMBERT'S DREAM

to leave the successive forms which the network takes on and focus our attention on the network alone.

MADemoiselle de l'Espinasse

Each strand of this sensitive network can be injured or tickled along its entire length. The pleasure or pain is here or there, in one location or another on one of the long legs of my spider, for I always come back to my spider. It's the spider which is located at the common origin of all the legs and which establishes that the pain or pleasure is at such and such a place without experiencing the pleasure or pain itself.

BORDEU

It's this continual, invariable interaction between all impressions and the common origin which constitutes the unity of the animal.

MADemoiselle de l'Espinasse

And it's the memory of all these successive impressions which creates for each animal the history of its life and of its individuality.

BORDEU

And it's the memory and the comparison which necessarily come from all these impressions which create thought and reasoning.

MADemoiselle de l'Espinasse

Where is this comparison made?

BORDEU

At the centre of the network.

MADemoiselle de l'Espinasse

What about the network itself?

BORDEU

It does not have in its centre any sense unique to it. It cannot see or hear, and it doesn't suffer. It is produced and fed. It arises from a soft, insensitive, and inert substance which serves as a pad on which it sits, listens, judges, and pronounces.

MADemoiselle de l'Espinasse

It suffers no pain.

BORDEU

No. With the lightest impression on the centre of the network it stops responding, and the animal falls into a death-like state. If

D'ALEMBERT'S DREAM

you make this impression stop, it returns to its functions, and the animal is reborn.

MADemoiselle de l'EsPInasse

How do you know that? Has anyone ever made a man die and be born again at will?

BORDEU

Yes.

MADemoiselle de l'EsPInasse

And how did that happen?

BORDEU

I'm going to tell you. It's a strange case. La Peyronie, whom you may have known, was called to visit an invalid who had received a violent blow on the head.²² The invalid felt his head beating. The surgeon was sure that an abscess had formed on the brain, and there wasn't a moment to lose. He shaved the invalid and opened the skull. The point of his instrument struck the very centre of the abscess. It had pus in it. He drained off the pus and cleaned the abscess with a syringe. When he pushed his injection into the abscess, the invalid closed his eyes, his limbs went immobile and inert, without the least sign of life. When he pulled back on the syringe and relieved the weight and pressure of the injected fluid on the centre of the network, the invalid reopened his eyes, moved, spoke, felt, was reborn, and came to life.

MADemoiselle de l'EsPInasse

That is remarkable. And did this invalid recover?

BORDEU

He did. And when he was cured, he reflected, thought, and reasoned. He had the same intelligence, the same good sense, the same ability to sort things out, with a good portion of his brain gone.

MADemoiselle de l'EsPInasse

That judge you referred to in the network is an extraordinary creature.

BORDEU

It sometimes makes mistakes. Its habits tend to make it biased—the way people feel pain in a limb which they don't have any more. You can mislead it whenever you like: if you

²²Francois de la Peyronie (1678-1747): a well-known and important French surgeon.

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cross your fingers one on top of the other and touch a small ball
it will declare there are two of them.

MADemoiselle de l'Espinasse

That shows it's just like all judges in the world and needs
experience, without which it will mistake the feeling of ice for
that of fire.

BORDEU

It does many other things. It gives an almost infinite volume to
an individual or shrinks the individual down almost to a point.

MADemoiselle de l'Espinasse

I don't understand you.

BORDEU

What is it that establishes a limit to your actual extent, the true
sphere of your sensibility?

MADemoiselle de l'Espinasse

My senses of sight and touch.

BORDEU

During the day, yes, but at night, in the shadows, above all when
you're dreaming about something abstract, or even during the
day when your mind is preoccupied.

MADemoiselle de l'Espinasse

Nothing. I exist as a point. I almost cease to be something
material. I feel only my thought. There is no more sense of place
or movement or body or distance or space for me. The universe
is annihilated for me, and I am nothing to it.

BORDEU

There you have the final limit to the concentration of your
existence, but, in theory, its expansion could have no limits.
When the true limit of your sensibility has been passed, whether
by moving into yourself, shrinking into yourself, or in extending
yourself outwards, we no longer know what that can become.

MADemoiselle de l'Espinasse

You're right, Doctor. Several times in dreaming it's seemed to
me . . .

BORDEU

And with invalids suffering an attack of gout . . .

D'ALEMBERT'S DREAM

MADemoiselle de l'Espinasse

. . . that I was getting huge . .

BORDEU

. . . their foot seemed to touch the canopy over the bed.

MADemoiselle de l'Espinasse

. . . that my arms and legs were getting infinitely longer and the rest of my body was acquiring a proportional volume, that Enceladus in the fable was nothing but a pigmy, that Amphitrite in Ovid, whose long arms went to make an immense belt around the earth, was nothing but a dwarf in comparison to me, and that I was mounting up into the sky and embracing both hemispheres.²³

BORDEU

Yes, that's really good. And I knew a woman in whom the phenomenon worked in reverse.

MADemoiselle de l'Espinasse

What! She got smaller gradually and shrunk into herself?

BORDEU

To the point where she felt herself as tiny as a needle. She saw, heard, reasoned, and judged, and she had a mortal fear of losing herself. She trembled when the smallest objects came near her. She didn't dare to budge from where she was.

MADemoiselle de l'Espinasse

Now, there's a strange dream—really distressing and inconvenient.

BORDEU

She wasn't dreaming at all. It was one of those things that happen when the menstrual cycle stops.

MADemoiselle de l'Espinasse

Did she remain long with this sense of being a tiny imperceptible woman?

BORDEU

One or two hours, after which she returned in stages to her natural size.

²³Enceladus in Greek mythology was one of the Giants, a huge monster. Amphitrite in Ovid is synonymous with the seas surrounding the land.

D'ALEMBERT'S DREAM

MADemoiselle de l'EsPInasse

What is the reason for these odd sensations?

BORDEU

In their natural and calm condition, the threads of the bundle have a certain tension, a tone, and a habitual energy which determines the real or imagined extent of the body. I say real or imagined, because since this tension, this tone, and this energy are variable, so our bodies are not always the same volume.

MADemoiselle de l'EsPInasse

And so in physical matters, just as with moral issues, we are subject to thinking ourselves greater than we are?

BORDEU

Cold makes us smaller, heat makes us larger, and an individual of a certain sort can believe all his life that he is smaller or larger than he really is. If it so happens that the mass of the bundle goes through a state of violent irritation, so the threads begin to stand up and the countless multitude of their extremities begin to push themselves out beyond their customary limits, then the head, feet, other limbs, and all the points on the surface of the body will be shifted an immense distance, and the individual will feel himself gigantic. The reverse phenomenon will occur if insensibility, apathy, and inertia take over the extremity of the threads and move gradually towards the centre of the bundle.

MADemoiselle de l'EsPInasse

I see that this expansion cannot be measured, and I also see that this insensibility, apathy, and inertia at the end of the threads, this numbness, after progressing some extent could establish itself and end up . . .

BORDEU

Just like what happened to La Condamine.²⁴ At that point the individual feels as if he has balloons under his feet.

MADemoiselle de l'EsPInasse

He exists beyond the limits of his physical sensations, and if he was enveloped in this apathy in every way, to us he'd seem like a small man living within a dead one.

BORDEU

From that you conclude that the animal which was nothing but

²⁴Charles de la Condamine (1701-1774) was a French explorer who spent years in South America. He suffered from deafness and paralysis before his death.

D'ALEMBERT'S DREAM

a point at its origin still doesn't know if it is, in reality, anything more. But let's go back.

MADemoiselle de L'ESPINASSE

Where to?

BORDEU

Where? To La Peyronie's trepanning. . . . There, I think, you had just what you asked me for, the example of a man who alternated between life and death. But there are better.

MADemoiselle de L'ESPINASSE

What could that be?

BORDEU

The myth of Castor and Pollux come to life—two children in whom the life of one was immediately followed by the death of the other, and the life of the latter immediately followed from the death of the first.²⁵

MADemoiselle de L'ESPINASSE

Now that's a fine story. Did that last a long time?

BORDEU

This existence lasted for two days which they shared equally, going through the different cycles, so that each one had one day of life and one day of death respectively.

MADemoiselle de L'ESPINASSE

I fear, Doctor, that you're abusing my credulity somewhat. Take care—if you deceive me once, I won't believe you anymore.

BORDEU

Do you sometimes read the *Gazette de France*?

MADemoiselle de L'ESPINASSE

Never, although it's a masterpiece by two intelligent men.²⁶

²⁵Castor and Pollux were twin brothers of Helen of Troy. Pollux was a son of Zeus and hence immortal. When Castor, the son of Tyndareus, died, Pollux asked Zeus if Castor could share his immortality. Zeus then let them alternate living on earth and in heaven, each one changing day by day. He also placed the two brothers in the constellations as Gemini.

²⁶The *Gazette de France* was a weekly newspaper controlled by the government. The two men referred to are Jean-Baptiste Suard (1732-1817) and Francois Arnaud (1721-1784).

D'ALEMBERT'S DREAM

BORDEU

See if someone will lend you this month's September 4 issue, and you'll see that in Rabastens in the diocese of Albi, two girls were born back to back, joined at their last lumbar vertebrae, their buttocks, and the hypogastric region. They couldn't hold one of them upright unless the other's head was down. When they were lying down, they could look at each other. Their thighs were bent between their trunks, and their limbs were raised. In the middle of the common circular line where they were attached in the hypogastric area, one could discern their sex, and between the right thigh of one and the corresponding left thigh of her sister, in a cavity there was a small anus through which meconium came out.²⁷

MADemoiselle de l'ESPINASSE

That's a really odd case.

BORDEU

They took in milk given to them on a spoon. As I told you, they lived for twelve hours, one losing consciousness as the other came out of unconsciousness, one dead while the other lived. The first blackout of one and the first life for the other was at four hours. The alternating blackouts and returns to life which came afterwards were shorter. They died at the same moment. People noticed that their navels also had an alternating movement outwards and inwards, going in for the one who was unconscious and going out for the one who was returning to life.

MADemoiselle de l'ESPINASSE

And what are do you say about these alternating periods of life and death?

BORDEU

Perhaps nothing worth much, but since we see everything through the spectacles of our own system and I don't want to be an exception to the rule, I say that it's the phenomenon of La Peyronie's trepanning but doubled in two joined beings. The networks in these two children were so thoroughly mixed together that they acted on and reacted to each other. When the centre of the bundle of one had the upper hand, it took control of the other child's network, and she immediately blacked out. And when the network of the second child dominated their common system, the situation reversed. In La Peyronie's trepanning patient, the pressure was directed downward from

²⁷The description of these twins appeared in the *Gazette de France*, September 4, 1769. Diderot's description of them here seems rather muddled and difficult to visualize precisely. Meconium is the baby infant's first feces.

D'ALEMBERT'S DREAM

above by the weight of a fluid; in the twin girls of Rabastens, the pressure came up from below through the tension in a certain number of strands in the network: this hypothesis is supported by the alternating inward and outward movement of their navels—in the one returning to life the navel came out, and in the one dying it went back in.

MADemoiselle de l'EsPInasse

There we have an example of two linked souls.

BORDEU

One animal based on the principle of two sensing systems and two areas of consciousness.

MADemoiselle de l'EsPInasse

But only having the use of one of them by itself at any given time. Still, who knows what would have happened if this animal had lived?

BORDEU

With the experience of all these moments of life and the most powerful habits one could imagine, what sort of intercommunication would have been established between these two brains?

MADemoiselle de l'EsPInasse

Double senses, a double memory, a double imagination, a double ability to focus—one half of a being which observes, reads, meditates, while its other half rests; then this other half takes up the same functions when its companion is weary: the double life of a double being.

BORDEU

That's possible. And with time nature brings out everything possible, so it will produce some strange compound creation.

MADemoiselle de l'EsPInasse

How impoverished we would be in comparison with such a being!

BORDEU

But why? There are already so many uncertainties, contradictions, and foolish things in a simple understanding that I have no idea anymore what would happen with a double understanding . . . But it is half past ten, and I hear a patient calling me from the suburbs.

D'ALEMBERT'S DREAM

MADemoiselle de L'ESPINASSE

Would he be in any real danger if you did not visit him?

BORDEU

Probably less than if I do visit. If nature can't do the work without me, then we'll have a good deal of trouble doing it together, and it's certain that I'll not get it done without her.

MADemoiselle de L'ESPINASSE

So why not remain?

D'ALEMBERT

Doctor, one word more, and I'll send you to your patient. Given all the changes I've been through in the course of my lifetime, I probably don't have now a single one of the molecules which I brought into the world when I was born. So how have I retained my identity for other people and for myself?

BORDEU

You told us that while you were dreaming.

D'ALEMBERT

Have I been dreaming?

MADemoiselle de L'ESPINASSE

All night long—and it was so much like a delirium I sent someone out to find the doctor this morning.

D'ALEMBERT

All that for the business of the spider's legs moving on their own, thus keeping the spider alert and making the animal talk. And what did the animal say?

BORDEU

It said that through its memory it retained its identity for others and for itself. And I'd add through the slowness of the changes. If you'd passed in the wink of an eye from youth to decrepitude, you'd have been thrown into this world as if at the first moment of your birth, and you'd have no longer been yourself either to others or to yourself, and other people would not have been themselves for you. All your interconnections would have been destroyed, the whole history of your life would have been jumbled up for me, and all the history of my life would have been jumbled up for you. How could you have known that this man, bent over his stick, with no spark in his eyes, dragging himself along with difficulty, still more strange to himself inside than on the outside, was the same man who yesterday was walking along so lightly, shifting quite heavy loads, who was able

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to give himself over to the most profound meditations, to the most delicate and the most powerful exercises? You wouldn't have understood your own works, you wouldn't have recognized yourself or anyone, and no one would have recognized you. The entire picture of the world would have changed. Remember that there was even less difference between you as a new born and a young person than there would be between you as a young man and you if you suddenly became a decrepit old man. Keep in mind that, although your birth was linked to your youth by a sequence of uninterrupted sensations, the first three years of your existence have never been in the history of your life. So what would the time of your youth have been for you if it had not been linked at all to the moment of your decrepitude? The decrepit D'Alembert would not have the slightest memory of the young D'Alembert.

MADemoiselle de l'EsPInasse

In the cluster of bees, there would not have been one who'd had the time to develop a sense of the larger group.

D'ALEMBERT

What are you talking about?

MADemoiselle de l'EsPInasse

I'm saying that the monastic spirit maintains itself because the monastery refurbishes itself gradually. When a new monk enters, he encounters a hundred old men who train him to think and feel as they do. A bee goes away. It is succeeded in the cluster by another who's soon up to date with what's going on.

D'ALEMBERT

Come on, what's all this extravagant talk of yours about monks, bees, clusters, and your monastery.

BORDEU

Not as extravagant as you might think. If there's only one consciousness in an animal, there are countless wills at work, for each organ has its own.

D'ALEMBERT

Why do you say that?

BORDEU

Well, I meant that the stomach wants food, but the palate does not. The difference between the palate or the stomach and the entire animal is that the animal knows what it wants, but the stomach and the palate have desires without knowing it. The stomach or the palate is to the complete animal almost like the

D'ALEMBERT'S DREAM

brute beast is to the human being. The bees lose their own consciousness but retain their appetites or desires. The fibre is a simple animal; the human being is a complex animal. But let's keep that issue for another occasion. To remove from a human being his consciousness of himself doesn't require sudden decrepitude—a considerably smaller event can do it. A man on the point of death receives the sacraments with a profound piety. He confesses his sins, asks his wife's forgiveness, kisses his children, summons his friends, speaks to his doctor, gives instructions to his servants, dictates his last wishes, puts his affairs in order—and all that with the soundest judgment, with his intelligence fully engaged. Then, once cured, he convalesces and hasn't the least idea of what he said or did during his illness. This period of time, sometimes very long, has disappeared from his life. There are even examples of people who have resumed the conversation or the action which the sudden attack of illness interrupted.

D'ALEMBERT

I remember that in a public academic exercise, a college pedant, all puffed up with his knowledge, was completely put down, as they say, by a Capucin whom he despised. He—put down completely! And by whom? By a Capucin! And what was the question under discussion? The contingent future—the *science moyenne* which he had been thinking about all his life!²⁸ And in what circumstances! In front of a crowded assembly, in front of his own students! There he was, his honour gone. His head worked over these ideas so much that he fell into a lethargic state which took from him all the knowledge he'd acquired.

MADemoiselle de l'ESPINASSE

But that was a fortunate thing for him.

D'ALEMBERT

I'd say you're right. He retained his good sense, but he'd forgotten everything. People taught him to speak and read again, and he died just when he was beginning to spell reasonably well. This man was no idiot. It was said he even had a certain eloquence.

²⁸*Science moyenne* is the French for the Latin term *scientia media*, taken from Medieval Catholic theology: "The future free actions of rational creatures which would have occurred if certain conditions had been fulfilled. Although they will not take place, they are nevertheless known by God in what is called the divine mediate knowledge, *scientia media*, i.e., a knowledge midway between God's foreknowledge of pure possibilities and of real futures that will come to pass" (from CatholicCulture.org). The Capuchins were an order of friars, devoted to a very simple and austere life.

D'ALEMBERT'S DREAM

MADemoiselle de l'Espinasse

Since the doctor has listened to your story, he has to listen to mine, too. A young man about eighteen or twenty years of age, whose name I don't remember . . .

BORDEU

He was M. de Schullemburg from Winterthur, and he was only fifteen or sixteen.

MADemoiselle de l'Espinasse

This young man suffered a fall in which he received a severe head concussion.

BORDEU

What are you calling a severe concussion? He fell off the top of a barn. His head was smashed in, and he remained unconscious for six weeks.

MADemoiselle de l'Espinasse

Whatever happened, do you know what followed this accident? The same as with your pedant. He forgot everything he knew. He was brought back as a very young child. He had a second childhood which lasted a while. He was timid and petty. He amused himself with toys. If he did something wrong and someone told him off, he went to hide in a corner. He asked if he could go to the bathroom for a number one or a number two. They taught him to read and write. But I forgot to tell you that it was necessary to teach him to walk again. He became a man once more—and a clever one, too. He left a work on natural history.

BORDEU

It was a series of engravings, plates to go with Mr. Sulzer's work on insects, following the system of Linnaeus.²⁹ I knew about this incident. It happened in the canton of Zurich, in Switzerland, and there are a number of similar examples. If you disturb the centre of the bundle, you change the animal. Apparently the animal has its whole being there, sometimes dominating the various branches and sometimes dominated by them.

MADemoiselle de l'Espinasse

The animal is subject to either despotism or anarchy.

²⁹Johann Sulzer (1735-1813) was a Swiss natural scientist who studied and classified insects. Carl Linnaeus (1707-1778): Swedish natural philosopher who established the basis for modern systems of classifying and naming in biology.

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BORDEU

Subject to despotism—that's a very good expression—the centre of the bundle issues its orders, and all the rest obeys. The animal is master of itself, *mentis compos* [*of sound mind*].

MADemoiselle de L'ESPINASSE

In a state of anarchy all the threads of the network rise up against their commander and there's no longer a supreme authority.

BORDEU

Exactly right. In great fits of passion, in delirium, or when danger is imminent, if the master directs all the forces of his subjects towards a single point, the feeblest animal manifests an incredible strength.

MADemoiselle de L'ESPINASSE

With fits of the vapours, there is a type of anarchy which is peculiar to us women.

BORDEU

It's the image of a weak administration, where each person arrogates to himself the authority of the master. I know only one way to cure the condition. It's difficult, but effective: the central part of this sensitive network, the part which makes up its identity, has the ability to be so affected by a powerful motive that it recovers its authority.

MADemoiselle de L'ESPINASSE

And then what happens?

BORDEU

What happens is that it does, in fact, regain its control, or the animal dies. If I had the time, I'd tell you two remarkable things about that.

MADemoiselle de L'ESPINASSE

But, Doctor, the time for your visit is past, and your patient isn't expecting you anymore.

BORDEU

I really shouldn't come here except when there's nothing to do, because it's just impossible to get away.

MADemoiselle de L'ESPINASSE

All right, so much for your moody outburst. Now what about your stories?

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BORDEU

Today you'll have to content yourself with this one. After her pregnancy, a woman fell into a most frightening condition of vapours—involuntary crying and laughter, asphyxiation, convulsions, swelling in the throat, gloomy silences, piercing cries—everything bad. This lasted for several years. She was passionately in love, and she believed she was seeing her lover, weary of her illness, beginning to grow distant. So she resolved to get better or die. A civil war developed in her, in which sometimes the master was victorious and sometimes the subjects. If the action of the threads in the network happened to be equal to the reaction from their centre, she fell down as if dead. She was taken to her bed, where she remained for whole hours without moving, almost lifeless. At other times, she got off with some lassitude and a general weakness, a wasting away which seemed it might be final. She remained six months in this state of war. The rebellion always began with the threads. She felt it coming on. At the first symptom, she'd get up, run around, devote herself to the most strenuous exercises—climbing up and down stairs, sawing wood, digging up the ground. The organ of her will power at the centre of her network got stronger. She used to say to herself, "Victory or death." After countless victories and defeats, the chief emerged the master, and the subjects became so submissive that, although this woman went through all sorts of domestic troubles and suffered different illnesses, there was no more trouble with vapours.

MADemoiselle de L'ESPINASSE

That was good, but I think I'd have done much the same.

BORDEU

That's because if you were in love, you'd love well, and because you're strong.

MADemoiselle de L'ESPINASSE

I understand. One is strong if by habit or by structure the centre of the network dominates the threads. On the other hand, one is weak if that centre is dominated by the threads.

BORDEU

We can derive many other conclusions from that.

MADemoiselle de L'ESPINASSE

But what about your other story? You can draw your conclusions later.

BORDEU

A young woman had devoted herself to several affairs. One day

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she made a decision to shut the door on her pleasures. There she was alone and melancholy and suffering from vapours. She had me summoned. I advised her to put on a peasant costume, dig up the ground all day, sleep on straw, and live on stale bread. This style of life did not please her. "So travel," I told her. She made a tour of Europe and recovered her health on the road.

MADemoiselle de L'ESPINASSE

That was not what you were going to say. But no matter—let's move on to your conclusions.

BORDEU

I'd never be able to finish.

MADemoiselle de L'ESPINASSE

So much the better. Tell me anyway.

BORDEU

I don't have the energy.

MADemoiselle de L'ESPINASSE

Why not?

BORDEU

Because at the rate we're going we'll just skim over everything and not go into anything in depth.

MADemoiselle de L'ESPINASSE

What does that matter? We're not composing anything—we're having a conversation.

BORDEU

For example, if the centre of this network calls back all the energies into itself, if the whole system, so to speak, moves backwards, as I believe happens in a man who meditates profoundly, in a fanatic who sees the skies open, in a savage who sings in the middle of the flames, in ecstasy, in voluntary or involuntary insanity . . .

MADemoiselle de L'ESPINASSE

Well then?

BORDEU

Well, then the animal becomes impassive. It's alive only at a single point. I didn't see the priest of Calamos whom St. Augustine talks about, the man who used to lose himself to the point of no longer feeling burning coals. And I didn't see that group of savages being tortured on the rack who smiled at their

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enemies, insulted them, and suggested to them more exquisite torments than those they were being made to suffer. I didn't see in the gladiatorial circus those gladiators who, as they were dying, remembered the grace and lessons of their gymnasium. But I believe all these cases, because I have seen—and seen with my own eyes—behaviour just as extraordinary as any of those.

MADemoiselle de L'ESPINASSE

Tell me about it, Doctor. I'm like a child—I love marvellous facts, and if they bring honour to the human race, I rarely challenge their credibility.

BORDEU

In the small town of Langres in Champagne, there was a good priest called Le Moni or De Moni, very devout and well versed in the truth of religion. He suffered attacks of a kidney stone, and it was necessary to operate on him. The day was chosen, the surgeon, his assistants, and I go to his house. He receives us with a serene expression. He undresses and lies down on his bed. We want to tie him down, but he refuses. "Just set me in place," he says, "in a convenient position." So we do that. Then he requests a large crucifix which was at the foot of the bed. We give it to him. He holds it between his arms and presses his mouth against it. We operate. He remains motionless. No tears or sighs escape him, and he was freed of the stone. He didn't know it had taken place.

MADemoiselle de L'ESPINASSE

That's a fine story. After that, you still doubt that the man whose ribs people broke with rocks saw the heavens open.

BORDEU

Do you know about ear aches?

MADemoiselle de L'ESPINASSE

No.

BORDEU

Lucky for you. It's the cruellest of all illnesses.

MADemoiselle de L'ESPINASSE

Worse than tooth ache—which unfortunately I do know something about?

BORDEU

There's no comparison. One of your friends, a philosopher, was tormented by one for two weeks. Then one morning he said to his wife, "I don't think I've enough endurance to get through

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the whole day. . . .” He thought that his only solution was to trick the pain artificially. Little by little he lost himself in thought about a question of metaphysics or geometry so completely that he forgot about his ear. They served him something to eat, and he dined without noticing his ear ache and reached the hour of his bedtime without suffering anything. The horrible pain did not come back until the tension in his mind stopped, but then it came with an incredible intensity, either because his weariness had irritated the sickness or because his weakness had made him less able to endure it.

MADemoiselle de l'Espinasse

In coming out of that condition, one must be really emotionally drained to the point of exhaustion. That's what sometimes happens to this man here.

BORDEU

That's dangerous. He should be careful.

MADemoiselle de l'Espinasse

I'm always telling him that, but he pays no attention.

BORDEU

That how he lives—he cannot control it any more. It will be the death of him.

MADemoiselle de l'Espinasse

That diagnosis makes me fearful.

BORDEU

What does this weariness, this lassitude, prove? That the threads in the network have not remained idle, and there's been an acute tension towards the common centre in the entire system.

MADemoiselle de l'Espinasse

What if this tendency toward acute tension lasts and becomes habitual?

BORDEU

That creates a spasm in the centre of the network. The animal goes mad—almost beyond any cure.

MADemoiselle de l'Espinasse

Why?

BORDEU

Well, a spasm in the centre is not like a spasm in one of the strands. The head is perfectly capable of commanding the feet,

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but the feet cannot command the head. The centre can command one of the strands, but the strand cannot command the centre.

MADemoiselle de l'Espinasse

Please explain the difference to me. In fact, why don't I think throughout my body? That's a question I should have thought about earlier.

BORDEU

It's because consciousness has only one location.

MADemoiselle de l'Espinasse

Well, there's a quick answer.

BORDEU

It can be in only one location, at the common centre of all sensations, the place where the memory sits and comparisons are made. Each thread is susceptible only to a certain fixed number of impressions, successive sensations, in isolation and unremembered. The centre is susceptible to everything. It is the registry of sensations. It retains the memory of them or of a sustained sensation, and the animal is led from its first formation to connect itself with this centre, to fix its entire identity there, and to exist in it.

MADemoiselle de l'Espinasse

What if my finger could have a memory?

BORDEU

Your finger would think.

MADemoiselle de l'Espinasse

So what then is memory?

BORDEU

It's the property of the centre, the specific sense of the centre of the network, just as sight is the property of the eye. And it's no more astonishing that the eye has no memory than that the ear has no sense of sight.

MADemoiselle de l'Espinasse

Doctor, you're evading my questions rather than dealing with them properly.

BORDEU

I'm not evading anything. I'm telling you what I know, and I would know more if I understood as much about the structure

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of the centre of the network as I do about the structure of its threads and if I'd had the same ability to observe it. But if I'm weak on certain specifics, I'm very strong on general phenomena.

MADemoiselle de L'ESPINASSE

And these general phenomena are?

BORDEU

Reason, judgment, imagination, madness, imbecility, ferocity, instinct.

MADemoiselle de L'ESPINASSE

I understand. All these qualities are only consequences of the original or habitually acquired relationship between the centre of the network and its branches.

BORDEU

Exactly. Is the principal part or trunk too vigorous in relation to the branches? That's how we get poets, artists, people with imagination, timid people, zealots, and fools. If it's too feeble? That gives us what we call brutes, ferocious animals. If the total system is lax and soft, without energy? That's how we get imbeciles. And if the whole system is energetic, harmonious, and well ordered? Well, then we get the good thinkers, philosophers, and wise men.

MADemoiselle de L'ESPINASSE

And depending on the all-powerful branch which dominates we get the different instincts in animals and the various special abilities in human beings—the sense of smell in dogs, the sense of hearing in fish, the sense of sight in the eagle, the talent for mathematics in D'Alembert, for mechanical things in Vaucanson, for music in Grétry, poetry in Voltaire, the different effects of one strand in the network being more energetic in them than any other and than similar strands in beings of their species.³⁰

BORDEU

And from habits which take control, as in the old man who loves women and Voltaire who still writes tragedies. [*At this point the doctor begins to daydream. Mademoiselle de L'Espinasse speaks to him*].

³⁰Jacques de Vaucanson (1709-1782) was a French inventor. André Grétry (1741-1813) was a Belgian-French musical composer.

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MADemoiselle de L'ESPINASSE

You're dreaming, doctor.

BORDEU

That's right.

MADemoiselle de L'ESPINASSE

What are you dreaming about?

BORDEU

About Voltaire.

MADemoiselle de L'ESPINASSE

And . . . ?

BORDEU

I was reflecting on the way great men are made.

MADemoiselle de L'ESPINASSE

How are they made?

BORDEU

How? Well, sensitivity

MADemoiselle de L'ESPINASSE

Sensitivity?

BORDEU

. . . or the extreme mobility of certain threads in the network is a dominant quality in mediocre creatures.

MADemoiselle de L'ESPINASSE

Ah, doctor, that's blasphemy!

BORDEU

I was expecting that reaction. But what is a sensitive being? A creature who's a slave to the wishes of his diaphragm. If a touching word strikes his ear or a remarkable sight strikes his eye, there he is all of a sudden caught up in an inner tumult. All the threads of his bundle are set in motion, a tremor runs through him, he's seized with horror, tears run down, sighs suffocate him, his voice breaks—the centre of the network has no idea what's going on. He has lost his composure, reason, judgment—all his resources.

MADemoiselle de L'ESPINASSE

That's a description of me.

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BORDEU

The great man who has the misfortune to receive this disposition from nature will spend all this time trying to weaken and dominate it, to make himself the master of his movements and to see that the centre of the network maintains all its imperial power. Then he'll retain his self-control in the midst of the greatest dangers, and he'll judge dispassionately but soundly. Nothing which can serve his point of view or contribute to his goal will escape him. People will have difficulty taking him by surprise. By the time he's forty-five years old, he'll be a great king, a great minister, a great politician, a great artist, above all a great actor, a great philosopher, a great poet, a great musician, a great doctor. He'll rule himself and everything around him. He'll have no fear of death, a fear which, as the Stoic philosopher has so sublimely stated, is a noose which the robust man seizes to lead the weak man wherever he wishes. He will have broken that noose and at the same time have freed himself from all the tyrannies of this world. Sensitive creatures or fools are on the stage, but he is in the orchestra seats. He's the wise man.

MADemoiselle de L'ESPINASSE

God preserve me from the company of such a wise man!

BORDEU

But because you haven't worked to be like him you'll go through an alternating series of acute pains and pleasures; you'll spend your life laughing and crying and will never be anything but a child.

MADemoiselle de L'ESPINASSE

I've resigned myself to that.

BORDEU

Are you hoping that will make you happier.

MADemoiselle de L'ESPINASSE

I've no idea.

BORDEU

Mademoiselle, this quality of sensitivity, which people prize so highly, never leads to anything great. It hardly ever manifests itself strongly without pain or weakly without boredom. One is either intoxicated or yawning. You surrender yourself without restraint to the sweet sensation of delicious music, or you let yourself be drawn in by the charm of a scene full of pathos. Your diaphragm is squeezed, the pleasure is past, and you've nothing left but a breathlessness which lasts all evening.

D'ALEMBERT'S DREAM

MADemoiselle de L'ESPINASSE

But what if I can enjoy sublime music and a stirring scene only on this condition.

BORDEU

That's a mistake. I also know how to enjoy or admire something, and I never suffer, unless from colic. I experience pure pleasure. That makes my criticisms much more stringent and my praise more gratifying and thoughtful. Is there such a thing as a bad tragedy for spirits as mobile as yours? How many times, as you read the work over, have you blushed at the emotional feelings you experienced in the theatre and vice versa?

MADemoiselle de L'ESPINASSE

Yes, that's happened to me.

BORDEU

So it's not appropriate for a sensitive being like you but rather for a tranquil and cool being like myself to say "This is true," "This is good," and "This is beautiful." . . . Let's strengthen the centre of the network. That's the best thing we can do. Do you know our life depends on that?

MADemoiselle de L'ESPINASSE

Our lives, Doctor! This is serious.

BORDEU

Yes, our lives. There's no one who hasn't at some time felt depressed. A single event is enough to bring on this involuntary and habitual feeling. And then, in spite of distractions, a variety of amusements, advice from friends, and one's own efforts, the threads stubbornly carry the gloomy impulses to the centre of the network. It's no good for the poor man to struggle—the spectacle of the universe keeps getting darker for him. He moves along with a cortege of mournful ideas which never leave him, and he finishes by killing himself.

MADemoiselle de L'ESPINASSE

Doctor, you make me afraid.

D'ALEMBERT: *[getting up, clothed in a dressing gown and a night cap]*

What about sleep, doctor? What do you say about that? It's something beneficial.

BORDEU

Sleep is the state where, whether through exhaustion or habit, the whole network relaxes and stays motionless. Then, as in sickness, each thread of the network is stimulated, moves, and

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transmits to the common centre a crowd of sensations which are often disparate, disjointed, and troubled. At other times they are so linked, so sequential, so well organized that the man would not have more reason or eloquence or imagination if he were awake. Then sometimes they are so violent and vivid, that the man, once awake, remains uncertain about the reality of things.

MADemoiselle de L'ESPINASSE

Well then, what is sleep?

BORDEU

It's a state in the animal where it is no longer a harmonious whole. All coordination and subordination stops. The master is left to the discretion of his servants and to the unbridled energy of his own activity. Is the optic thread stimulated? The centre of the network sees. And it hears if the auditory thread prompts it. Actions and reactions are the only things which remain between them. It's a consequence of the characteristics of the centre, of the law of continuity and of habit. If the action begins with the thread for sexual pleasure, the one which nature has fixed for erotic love and the propagation of the species, the effect of the reaction at the centre of the network will be the reawakened image of the loved one. But if, by contrast, this image is aroused first at the centre of the network the reaction will lead to tension in the thread for sexual pleasure, erection, and then ejaculation of seminal fluid.

D'ALEMBERT

And so there's a dream as we rise and a dream as we go down. I had one of the former last night, but I don't know where it went.

BORDEU

While we're awake, the network obeys the impressions of external objects. In sleep, it's the exercise of its own sensibility which gives rise to everything that takes place in it. There's no distraction in a dream—that's why it's so life like. It's almost always the result of some abnormal excitement, a temporary fit of illness. The centre of the network is alternately active and passive in an infinity of ways. That's where its disorder arises. Its concepts are sometimes as linked and distinct as in the animal confronting a natural spectacle. It's only the portrait of this spectacle reawakened. And indeed that's why it seems true and why it's impossible to distinguish it from the state of being awake. There's no probability that it's more one of these states than the other, no way of recognizing the mistake, other than experience.

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MADemoiselle de L'ESPINASSE

Is experience always able to do that?

BORDEU

No.

MADemoiselle de L'ESPINASSE

If a dream presents me with the picture of a friend whom I've lost and gives him to me as truly as if this friend were alive, if he speaks to me and I hear him, if I touch him and my hands get the impression of solidity from him, and if, when I wake up, my soul is full of tenderness and sorrow and my eyes full of tears and my arms are still stretched out towards the place where he appeared to me, what will convince me that I haven't really seen, heard, and touched him?

BORDEU

His absence. But if it is impossible to distinguish being awake from being asleep, who can appreciate how long sleep lasts? When it's peaceful, it's an unconscious interval between the moment of going to bed and the moment of getting up. When it's disturbed, it sometimes lasts for years. In the first case, at least, consciousness of self completely ceases. Can you give me a dream which no one has ever had and no one ever will have?

MADemoiselle de L'ESPINASSE

Yes—that one is someone else.

D'ALEMBERT

And in the second case, we not only have consciousness of ourselves but even of our will and liberty. What is this will, what is this freedom of the man who's dreaming?

BORDEU

What is it? It's the same as it is in the man who's awake—the last impression of desire or aversion, the final result of everything which one has been since birth right up to the moment where one is. And I defy the most nimble mind to notice the least difference between them.

D'ALEMBERT

You think so?

BORDEU

What a question, coming from you! You who, delivered over to profound speculations, have passed two-thirds of your life dreaming with your eyes open and acting without willing—yes, without willing, much less willing than in your dreams. In your

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dreams you commanded, gave orders, people obeyed you. You were unhappy or satisfied, experienced contradictions, encountered obstacles, got annoyed, loved, hated, blamed, approved, laughed, cried, came, and went. In the course of your meditations, your eyes were hardly open in the morning before you were caught up once more with the idea which had occupied you the previous day. You got dressed, sat at your table, meditated, drew diagrams, carried out calculations, dined, took up your calculations again, sometimes leaving your desk to confirm them. You talked to others, gave orders to your servants, dined, went to bed, and then to sleep without having done the least willed act. You were nothing but a point—you acted, but you didn't will. Does one have a will all by oneself? The will is always born from some interior or exterior motive, some present impression, some reminiscence of the past, some passion, some future project. After that I'll say only one word to you about liberty—it's the fact that the last of our actions is the necessary effect of a single cause—ourselves—a very complex cause, but a single one.

MADemoiselle de L'ESPINASSE

A necessary effect?

BORDEU

Undoubtedly. Try to imagine the production of a different action, assuming that the person acting is the same.

MADemoiselle de L'ESPINASSE

He's right. Since I act in this way, anyone who could act different is no longer me. And to assume that at the moment when I do or say something, I could say or do something different from that is to assume that I am myself and I am someone else. But, Doctor, what about vice and virtue? Virtue, such a holy word in all languages, such a sacred idea among all nations!

BORDEU

It's necessary to transform it into the idea of doing good and its opposite, doing harm. People are born fortunate or unfortunate and are imperceptibly led along by the general torrent which leads one to glory and the other to ignominy.

MADemoiselle de L'ESPINASSE

And what about self-esteem, shame, and remorse?

BORDEU

Puerile ideas based upon ignorance and vanity in a creature who

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credits himself with the merit or demerit which comes from an inevitable moment.

MADemoiselle de l'EsPInasse

And rewards and punishments?

BORDEU

Those are means to correct a modifiable creature we call bad and to encourage those we call good.

MADemoiselle de l'EsPInasse

In all this doctrine isn't there something dangerous?

BORDEU

Is it true or false?

MADemoiselle de l'EsPInasse

I believe it's true.

BORDEU

So you're saying you think that lies have their advantages and truth its disadvantages.

MADemoiselle de l'EsPInasse

That's what I think.

BORDEU

So do I. But the advantages of a lie are temporary and those of the truth are eternal. The detrimental consequences of the truth, when it has any, quickly pass, and those of lies don't end until the lie does. Examine the effects of a lie in a man's head and its effect on his conduct: in his mind the lie is more or less confused with the truth and his head has trouble reasoning, or else he's happy to incorporate the lie and his mind is simply mistaken. Now, what behaviour can you expect from a mind which is either inconsistent in its reasoning or logically consistent in its errors?

MADemoiselle de l'EsPInasse

The second of these vices may be less contemptible than the first, but is perhaps more to be feared than the first.

D'ALEMBERT

Excellent. See how everything comes back to sensitivity, memory, and organic movements. That suits me fine. But imagination, abstractions?

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BORDEU

Imagination . . .

MADemoiselle de L'ESPINASSE

One moment, doctor, let's summarize. According to your principles, it seems to me that by a sequence of purely mechanical operations, I could reduce the finest genius in the world to a mass of unorganized flesh to which we would not ascribe anything but sensitivity at a particular moment, and then we could bring back this unformed mass from a state of the most profound stupidity one could imagine to the condition of a man of genius. The first of these two phenomena would consist of mutilating a certain number of threads in the primitive tangle and really mixing the other ones up. The reverse phenomenon would require us to restore to the tangle the threads we had detached and to leave the whole thing to develop properly. For example, if I remove from Newton the two auditory threads, he has no more sense of sound, the olfactory threads, no more sense of smell, the optic threads, no more sense of colour, the threads of the palate, no more sense of taste—if I cut out or mix up the others, then farewell to the organic structure of the brain, farewell memory, judgement, desires, aversions, passions, willing, consciousness of the self, and lo and behold there is an unformed mass which retains nothing but life and sensitivity.

BORDEU

Two almost identical qualities. Life is the aggregate, and sensitivity is among the elements.

MADemoiselle de L'ESPINASSE

I take this mass again, I restore its olfactory threads, and its nose starts to work, the auditory threads and it hears, the optic threads and it sees, the palate threads, and it tastes. By straightening out the rest of the tangle, I permit the other strands to develop, and I see memory, comparison, judgment, reason, desires, aversions, passions, natural aptitude, and talent reborn. I recover my man of genius and all that without the intervention of any heterogeneous unintelligible agent.

BORDEU

That's exactly it. Stick to that. The rest is nothing but nonsense. But what about abstractions and the imagination? The imagination is the memory of forms and colours. The spectacle of a scene or an object necessarily sets up the sensing instrument in a certain manner. It either winds itself up on its own or is wound up by some foreign cause. Then it quivers inside or it makes some external sound. It either records in

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silence the impressions which it has received or it makes them burst out in conventional sounds.

D'ALEMBERT

But its account exaggerates, omits circumstances, adds things, distorts the facts or embellishes them, and the sensing instruments nearby imagine impressions which are really those of the resonating instrument and not those from event which has taken place.

BORDEU

That's true. The account can be historical or poetical.

D'ALEMBERT

But how is this poetry or falsehood introduced into the account?

BORDEU

By ideas which arouse each other, and they do so because they have always been linked. If you've taken the liberty of comparing an animal to a harpsichord, you'll allow me to compare a poetry recital to a song.

D'ALEMBERT

That's fair.

BORDEU

In every tune there is a scale. This scale has its intervals, each of its strings has its harmonics, and these harmonics have their own harmonics. In this way, modulations are introduced into passages of the melody, and the music is embellished and extended. The musical event is an established theme which each musician responds to in his own way.

MADemoiselle de L'ESPINASSE

But why complicate the question with this metaphorical style? I would say that since everyone has his own eyes, he sees differently and gives a different account. I'd say that each idea awakens others in a person and that, according to how his mind works or his character, a person either holds to ideas which represent the facts rigorously or else he introduces into them other ideas which have been aroused. I'd say that there's a choice to be made among these ideas. I'd say . . . well, that this subject alone, if we dealt with it thoroughly, would fill a book.

D'ALEMBERT

You're right. But that won't prevent me from asking the doctor if he is really persuaded that some shape which didn't look like

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anything could ever be produced in the imagination, a shape which would not be produced in a story.

BORDEU

I think that's possible. Every delirium of this faculty is basically like the talent of those charlatans who cut up several animals and then make up a strange one from the pieces, something we've never seen in nature.

D'ALEMBERT

What about abstractions?

BORDEU

There aren't any. There are only habitual omissions, ellipses which make propositions more general and language faster and more convenient. They are the linguistic signs which have given birth to the abstract sciences. A quality common to several actions gave rise to the words vice and virtue, a quality common to several beings gave rise to the words ugliness and beauty. People said one man, one horse, two animals, and then later they said one, two, three, and the whole science of numbers was born. We have no mental image of an abstract word. We have observed in all three-dimensional bodies length, width, and depth. We have busied ourselves with each of these dimensions, and from that we have derived all the mathematical sciences. All abstraction is nothing but a sign empty of an idea. All abstract science is only a combination of signs. We have excluded the mental image once we separated the sign from the physical object, and it's only by re-attaching the sign to the physical object that science becomes once again a science of ideas. That's where the need arises—so frequent in conversation and our written works—of dealing with examples. When, after a long combination of signs, you ask for an example, you are only asking the person talking to give body, form, reality, and some idea to the series of his verbal noises by linking them to some established sensations.

D'ALEMBERT

Is that clear enough for you, Mademoiselle?

MADemoisELLE DE L'ESPINASSE

Not all that much, but the doctor is going to explain himself.

BORDEU

It pleases you to say that. It's not that there isn't perhaps something to correct and plenty to add to what I've said, but it's half past eleven, and I have an appointment in the Marais at noon.

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D'ALEMBERT

A faster and more convenient language! Doctor, do we understand ourselves? Are we understood?

BORDEU

Almost all conversations are settled accounts . . . I don't know where my cane is any more . . . We have no idea present in our minds . . . And my hat . . . And for this sole reason—no man is perfectly like another one. We never understand exactly. We are never exactly understood. With everything it's a matter of more or less. Our speech always falls short of a sensation or goes beyond it. We clearly see the diversity in judgments, and there is a thousand times more we do not see and which fortunately we cannot see. Good bye, good bye . . .

MADemoiselle de L'ESPINASSE

One word more, please.

BORDEU

Speak quickly then.

MADemoiselle de L'ESPINASSE

You remember those jumps you spoke to me about?

BORDEU

Yes.

MADemoiselle de L'ESPINASSE: Do you think that idiots and intelligent men have jumps like that in their ancestry?

BORDEU

Why not?

MADemoiselle de L'ESPINASSE

So much the better for our grandchildren. Perhaps another Henry IV will reappear.³¹

BORDEU

Perhaps he's already come back.

MADemoiselle de L'ESPINASSE

Doctor, you'll come dine with us.

³¹Henry IV (1553-1610) was an extremely successful and popular king of France, noted for his religious tolerance.

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BORDEU

I'll do what I can. I'm not promising. If I can come, you'll be seeing me.

MADemoiselle DE L'ESPINASSE

We'll wait for you until two o'clock.

BORDEU

Agreed.

Denis Diderot
SEQUEL TO THE PRECEDING CONVERSATION

Speakers: Mademoiselle de L'Espinasse, Doctor Bordeu

[Around two o'clock the doctor returns. D'Alembert has gone out to dine, and the doctor finds himself in an intimate conversation with Mademoiselle de L'Espinasse. Dinner is served. They talk about fairly inconsequential things until dessert, but when the servants leave, Mademoiselle de L'Espinasse speaks to the doctor.]

MADemoisELLE DE L'ESPINASSE

Come, Doctor, drink a glass of Malaga, and then you can answer a question which has gone through my head a hundred times and which I wouldn't dare to ask anyone except you.

BORDEU

This is excellent Malaga. What's this question of yours?

MADemoisELLE DE L'ESPINASSE

What do you think of breeding between species?

BORDEU

On my word, that's another fine question! I think human beings have placed a great deal of importance on the reproductive act and they've been right to do so, but I'm not happy about their laws, both civil and religious.

MADemoisELLE DE L'ESPINASSE

And what do you find to complain about with them?

BORDEU

They were made without equity, purpose, or any regard for the nature of things and public utility.

MADemoisELLE DE L'ESPINASSE

Please try to explain yourself.

BORDEU

That's what I propose . . . But wait. *[He looks at his watch]* I still have a good hour to give you. I'll go quickly, so that we'll have enough time. We're alone. You're no prude, so you won't think that I wish to forget about respecting you the way I should. And however you judge my ideas, I hope for my part that you'll not conclude anything from them critical of my morality.

MADemoisELLE DE L'ESPINASSE

Certainly not. But your opening remarks disturb me.

SEQUEL TO THE PRECEDING CONVERSATION

BORDEU

In that case, let's change the subject.

MADemoiselle de L'ESPINASSE

No, no. Go on with what you were going to say. One of your friends who was looking for husbands for me and my two sisters gave a sylph to the youngest, a large angel of the Annunciation to the oldest, and a disciple of Diogenes to me. He understood all three of us well. Still, Doctor, you can veil some things, just a little.

BORDEU

Of course I will, as much as the subject and my profession permit.

MADemoiselle de L'ESPINASSE

That won't be very difficult for you. But here's your coffee. Drink it.

BORDEU [*After drinking his coffee*]

Your question is a matter of physical science, morality, and poetics.

MADemoiselle de L'ESPINASSE

Poetics!

BORDEU

Undoubtedly. The art of creating beings which don't exist by imitating those which do is true poetry. On this occasion, instead of citing Hippocrates, will you allow me to quote Horace? This poet, or maker, says somewhere *Omne punctum qui miscuit utile dulci*—the supreme merit is to have united the pleasant with the useful. Perfection consists in reconciling these two things. An action which is agreeable and useful should occupy first place in the aesthetic order, and we can't refuse second place to what is useful, so third place will go to what is pleasant. We'll give lowest rank to something which brings neither pleasure nor profit.

MADemoiselle de L'ESPINASSE

Up to this point I can share your opinion without blushing. Where is that going to lead us?

BORDEU

You'll see. Mademoiselle, can you inform me what profit or pleasure is produced by chastity and strict continence, either to the individual who practises them or to society?

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MADemoisELLE DE L'ESPINASSE

I'd say none at all.

BORDEU

Thus, despite the magnificent elegies which fanaticism has produced on their behalf and despite the civil laws which protect them, we'll delete them from the catalogue of virtues, and we'll agree that there is nothing so childish, ridiculous, absurd, harmful, and contemptible, nothing worse than these two rare qualities, with the exception of positive evil.

MADemoisELLE DE L'ESPINASSE

We can agree on that.

BORDEU

Be careful here. I warn you, you'll be backing away from this position in a moment.

MADemoisELLE DE L'ESPINASSE

No, I never retreat.

BORDEU

What about solitary sexual acts?

MADemoisELLE DE L'ESPINASSE

What about them?

BORDEU

Well, at least they bring pleasure to the individual. So either our principle is false or else . . .

MADemoisELLE DE L'ESPINASSE

Come now, doctor!

BORDEU

Yes, Mademoiselle, yes because such acts are just as harmless but they're not pointless. It's a need, and even if a person isn't drawn to them by need, it's always something pleasant. I want people to be healthy—that's my first priority. Do you understand that? I criticize all excess, but in a social condition like ours, for every reasonable consideration there are a hundred more (not including our emotional temperament and the dreadful consequences of a rigorous continence, above all with young people)—lack of money, men's fear of bitter regrets, a woman's fear of dishonour, things which reduce an unhappy creature dying of languor and boredom, a poor devil who doesn't know where to turn, to finish himself off in the manner of a Cynic. Cato told a young man on the point of entering a

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prostitute's house, "Courage, my son. . . ." Would he give him the same advice today? On the other hand, if he caught him alone in the act of masturbating would he not claim that that was better than corrupting another man's wife or risking his own honour and health? What then? Because circumstances do not permit me the greatest imaginable happiness, that of mixing my senses, intoxication, and soul with the senses, intoxication, and soul of a companion chosen by my heart and of reproducing myself in her and with her, because I cannot consecrate my action with the seal of utility, am I to forbid myself a necessary and sweet moment? We have our blood let when we suffer from a plethora, and what does it matter what the nature of the superabundant humour is, its colour, and the manner we relieve ourselves of it? It's just as superfluous in one of these conditions as in another. And if it's pumped out of its reservoirs and distributed throughout the machine, it evacuates itself by another longer route, more painful and dangerous. So won't it be lost anyway? Nature doesn't permit anything useless. And how could helping her make me guilty when she calls for my help with the most unambiguous symptoms? We should never provoke nature, but we can lend her a hand occasionally. Refusing to do that or just doing nothing I see as mere foolishness and the loss of pleasure. Live temperately, people tell me, wear yourself out with physical exhaustion. I understand what they're saying—I should deprive myself of one pleasure and go to great trouble to distance myself from another one. What an excellent idea!

MADemoiselle de l'EsPInasse

Now there's a doctrine which isn't a good one to preach to children.

BORDEU

Or to others. However, will you allow me an assumption? You have a good daughter, too good, and innocent, too innocent. She is at an age where her emotional life is developing. Her head is all upset, and nature is no help. You call me in. I see right away that all the symptoms which have you afraid are born from the superabundance and retention of sexual fluid. I warn you that she is threatened by a madness which is easy to prevent and which sometimes is impossible to cure. I indicate the remedy to you. What do you do?

MADemoiselle de l'EsPInasse

To tell you the truth, I think . . . but this example never happens.

BORDEU

Don't believe that. It's not rare, and it would be frequent if the

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looseness in our morals didn't stop it. . . . Whatever the case, to reveal these principles would be treading all decency underfoot, drawing all the most hateful suspicions onto oneself, and committing a crime against society. You're day-dreaming.

MADemoisELLE DE L'ESPINASSE

Yes, I was hesitating about asking you if it had ever happened that you'd had to give advice like that to any mothers.

BORDEU

Of course I have.

MADemoisELLE DE L'ESPINASSE

And what view did these mothers take?

BORDEU

All of them, without exception, chose the good side, the sensible side . . . I wouldn't raise my hat in the street to the man suspected of practising my doctrine. It would be fine with me if people called him despicable. But we are chatting without witnesses and without any consequences, and I'll tell you about my philosophy what a completely naked Diogenes said to the young and prudish Athenian against whom he was intending to compete in wrestling, "My son, don't be afraid of anything. I'm not as bad as that man over there."

MADemoisELLE DE L'ESPINASSE [*covering her eyes*]

Doctor, I see where you are going, and I bet you. . . .

BORDEU

I don't bet. You'll win. Yes, Mademoiselle, that's what I think.

MADemoisELLE DE L'ESPINASSE

What! You don't care whether people confine their sexual activity to those like themselves or not?

BORDEU

That's right.

MADemoisELLE DE L'ESPINASSE

You really are a monster.

BORDEU

It's not me—it's either nature or society. Listen, Mademoiselle. I don't allow myself to be limited by the words I use, and I explain myself all the more freely because I speak my mind and because the well-known purity of my morals doesn't expose me to criticism from any quarter. So I'll ask you this: if there are

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two actions equally concerned with and limited to sexual pleasure, both of which can give pleasure without utility, and if one of them gives pleasure only to the person who does it and the other shares the pleasure with a similar being, male or female, for here the gender of the participants or even the nature of the sexual acts is irrelevant, what would common sense say in favour of one or the other?

MADemoiselle de L'ESPINASSE

Questions like that are too lofty for me.

BORDEU

Ah, after being a man for four minutes, there you go taking up your wimple and petticoats again and becoming a woman once more. Well, all right, then. We must treat you as such. . . . That's that . . . No one says a word any more about Madame du Barry³². . . You see, everything works out. We thought that the court would be thrown upside down. But the master has acted like a sensible man: *Omne tulit punctum* [*He's won every point*]. He kept the woman who gives him pleasure and the minister who's useful to him . . . But you're not listening to me . . . Where's your head been?

MADemoiselle de L'ESPINASSE

I was thinking of those sexual combinations which all seem to me to be contrary to nature.

BORDEU

Everything which exists cannot be either against nature or outside of nature, and I don't even exclude chastity and voluntary continence, which would be most important crimes against nature if it was possible to sin against nature, and the most important crimes against the social laws of a country where actions were weighed in some scale other than fanaticism and prejudice.

MADemoiselle de L'ESPINASSE

I come back to your accursed syllogisms, and I don't see any middle ground. One must either deny everything or accept everything . . . But what about this, Doctor—the fairest and shortest way would be to jump over the quagmire and to get back to my first question: What do you think about inter-species breeding?

³²Madame du Barry (1743-1793) was an official mistress of Louis XV, king of France.

SEQUEL TO THE PRECEDING CONVERSATION

BORDEU

We don't need to jump to get there—we were already there.
Now, is your question about physical science or morality?

MADemoiselle de L'ESPINASSE

About science, physical science.

BORDEU

So much the better. The question of morality comes first, and
you have settled it. So then . . .

MADemoiselle de L'ESPINASSE

I agree . . . It's undoubtedly what has to come first, but I
wanted . . . you to separate cause and effect. Let's leave the ugly
cause out of it.

BORDEU

That's asking me to begin at the ending. But since that's what
you want, then I'll tell you that, thanks to our timidity, our
aversions, our laws, and our prejudices, there have been very
few experiments tried, and we don't know which sexual unions
would be completely sterile or cases where the useful would
combine with the pleasant or what sort of species we could
expect from various prolonged attempts. We don't know
whether fauns are real or fabulous, or whether we could
multiply the race of mules in a hundred different ways, or
whether those we do know about are truly sterile. But here's a
strange fact which countless educated people will tell you is true
but which is false, namely, that in the poultry yard of the
archduke they've seen a disgusting rabbit who used act as a cock
and service around twenty shameless hens who were happy with
the arrangement. The people will add that they have been
shown some chickens covered with fur, which were the
products of this bestiality. You can be sure they were ridiculed.

MADemoiselle de L'ESPINASSE

But what do you mean by some prolonged attempts?

BORDEU

I mean that the transitions from one creature to another are
gradual, that assimilations of beings need to be prepared for,
and that in order to succeed in these kinds of experiments, it
would be necessary to work for a long time at first bringing the
animals closer together by a similar diet.

MADemoiselle de L'ESPINASSE

It would be difficult to reduce a human being to grazing.

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BORDEU

But not to drink goat's milk frequently, and we could easily lead the goat to feed on bread. I've chosen the goat for my own particular reasons.

MADemoiselle de L'ESPINASSE

What reasons?

BORDEU

You are being very bold! Well, the reason is . . . it's because from them we would derive a vigorous race—intelligent, tireless, and quick. From them we could make excellent household servants.

MADemoiselle de L'ESPINASSE

A very good idea, Doctor. It seems to me that I can already see behind our duchesses' carriages five or six large insolent types with goats' feet, and that makes me happy.

BORDEU

And we wouldn't be degrading our brothers any more by subjecting them to work unworthy of them and us.

MADemoiselle de L'ESPINASSE

Even better.

BORDEU

And in our colonies we wouldn't be reducing human beings any more to the condition of beasts of burden.

MADemoiselle de L'ESPINASSE

Then quickly, Doctor, don't waste any time. Make us some goat-men.

BORDEU

And you'd allow that without any scruples?

MADemoiselle de L'ESPINASSE

Just a moment—I've thought of one. Your goat men would be dreadfully dissolute sexually.

BORDEU

I'm not guaranteeing that they'd have good morals.

MADemoiselle de L'ESPINASSE

There'd no longer be any safety for decent women. They'd multiply without end. And in time we'd have to knock them

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down or obey them. I don't want them any more—no, no more.
You can stop worrying.

BORDEU: [*as he goes to leave*] What about the question of their baptism?

MADemoiselle de L'ESPINASSE

That would make a fine fuss at the Sorbonne.

BORDEU

Have you seen in the King's Garden zoo, in a glass cage, that orang-utan who looks like a Saint John preaching in the desert?

MADemoiselle de L'ESPINASSE

Yes, I've seen it.

BORDEU

Well, Cardinal de Polignac said to it one day, "If you speak, I'll baptize you."

MADemoiselle de L'ESPINASSE

Good bye then, Doctor. Don't neglect us for ages, as you usually do, and remember sometimes that I love you madly. If people only knew all these horror stories you've been telling me.

BORDEU

I'm very sure you'll keep quiet about them.

MADemoiselle de L'ESPINASSE

Don't be too sure. I only listen for the pleasure of repeating them. But one word more, and for the life of me I won't return to the subject.

BORDEU

What's that?

MADemoiselle de L'ESPINASSE

What about these abominable tastes? Where do they come from?

BORDEU

Always from some deficiency in the organic structure of young people and a corruption of the minds of old men, from the attractions of beauty in Athens, the shortage of women in Rome, and the fear of the pox in Paris. Good bye, good bye.